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# AlphaCALC®
# Reference Book

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*AlphaCALC Reference Book*

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This document applies to AlphaCALC versions 2.0 and later.

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# PREFACE

This *Reference Book* is designed to give you the information you need in your everyday use of AlphaCALC. It has three major sections, covering the following areas:

If you're just starting to learn AlphaCALC, we suggest you read the *AlphaCALC Tutorial* before referring to this book. The *Tutorial* is a step-by-step guide to the basics of AlphaCALC.

- Part One includes general information about AlphaCALC; it covers how to call up AlphaCALC, use its menus and commands, see help information, and leave AlphaCALC again. In addition, it discusses entering information in your work-sheet: how to enter a cell formula, the different types of formulas and values, how to format values for the appearance you want, and so on.

- Part Two describes each of AlphaCALC's command and function key features, in alphabetical order.

- Part Three covers AlphaCALC's functions, again in alphabetical order. A function is a pre-programmed calculation you can use in a cell formula. AlphaCALC has functions to add numbers, compute averages, figure depreciation amounts, and many more—almost 100 in all.

In addition, there are four appendices:

- Appendix A is a quick reference alphabetic list of all commands and key features, with one sentence descriptions.

- Appendix B is a similar list for functions—one sentence descriptions for quick reference.

- Appendix C lists and explains the error and other informational messages you may see while using AlphaCALC.

- Appendix D lists the universal key sequences for all AlphaCALC function key and special key features.

## THE AlphaCALC DOCUMENTATION LIBRARY

This book is your everyday guide to using AlphaCALC. If you need more advanced or detailed information in some area, you can refer to these other documents:

- The *AlphaCALC Tutorial* is a beginner's introduction to AlphaCALC. If you've never used AlphaCALC, this is the easiest and fastest way to get started.
- The *Release Notes* for your version of AlphaCALC tell you what's new in your version of AlphaCALC. They also tell you what versions of other Alpha Micro products are compatible with your version of AlphaCALC.
- The *AlphaCALC Installation and Administrator's Guide* gives complete instructions on setting up AlphaCALC on your system. It also includes information of interest to the system administrator: how to set AlphaCALC's operating options, how to change AlphaCALC help files and other messages, and so on.
- *AlphaCALC Programmer's Information* describes AlphaCALC's more technical aspects. It covers sharing AlphaCALC's data with other spreadsheets or your own software and the internal formats in which AlphaCALC stores information.
- *AlphaCALC Keyboard Commands Quick Reference* lists the keys or key sequences to use to activate all of AlphaCALC's key features, depending on the type of terminal you're using.

## GRAPHIC CONVENTIONS

In order to make this document as easy as possible to understand, we use several conventions in layout and type style, so you can recognize whether we are talking about a command or key, when you're to type an entry, and so on.

| CONVENTION | EXPLANATION |
|---|---|
| FILE.EXT | AMOS command names and file names, such as CALC.INI, are shown in all capitals. |
| AlphaCALC Main Menu | AlphaCALC screen names are printed in the same type face as regular text, with initial capitals. |
| **`What You Type`** | Text you are expected to type is shown in a bold Courier typeface. |
| `What AlphaCALC Displays` | Text AMOS or AlphaCALC displays on your screen is shown in Courier typeface. |
| **PRINT** | AlphaCALC command names are shown in small, bold capitals. |

| CONVENTION | EXPLANATION |
|---|---|
| KEY | Keys you need to press are shown in a keycap box. |
| KEY 1 / KEY 2 | Two keys separated by a slash, like this, means you need to press them at the same time. Hold down the first key and press the second one. |
| FUNCTION | AlphaCALC function names are in all capitals. |
| `NAME(`*`argument`*`)` | Function formats are given in Courier typeface, with variable parts of the argument in italics. |
| | The halt symbol marks important information you should read carefully before continuing. This often indicates something you MUST or MUST NOT do, so read carefully. |
| | The hint symbol marks a helpful bit of information. It often points out a short cut which may save you time or trouble. |
| | The remember symbol marks information you need to keep in mind while you are following a set of instructions. |

# PART ONE
# GENERAL INFORMATION

Part 1 of this book gives you general information, applicable throughout AlphaCALC. The topics covered include:

- Starting and leaving AlphaCALC.
- AlphaCALC's design.
- How to choose AlphaCALC commands.
- Using command screens.
- Using key features
- Viewing help information.
- The worksheet display.
- Moving around the worksheet.
- Working with cell IDs and cell ranges.
- Entering information in the worksheet.
- Cell formulas and cell values.
- Types of cell formulas.
- Cell formats.
- How AlphaCALC displays information.
- Color in AlphaCALC.

## STARTING AlphaCALC

To start using AlphaCALC, at AMOS command level, type:

**CALC** [RETURN]

This takes you to the AlphaCALC Main Menu. From there, you can create a new worksheet, edit an existing one, or use any other command listed.

You can edit a worksheet directly from AMOS by including the worksheet name in the CALC command. For example, to go directly to the worksheet SAMPLE, type:

**CALC SAMPLE** [RETURN]

If you have two versions of AlphaCALC on your computer, the command you use may be something other than CALC. Also, depending on how your computer is set up, you may choose AlphaCALC from a menu rather than entering the CALC command.

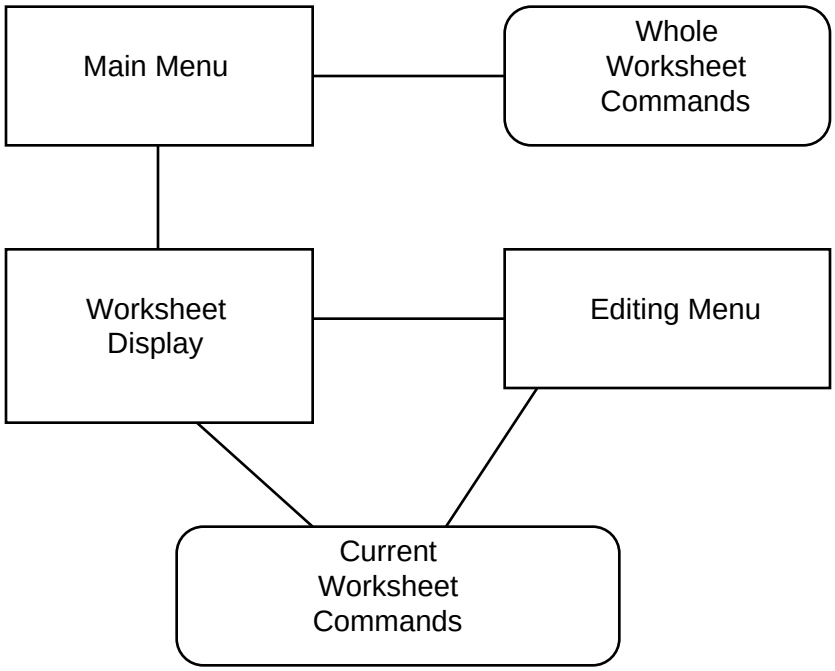
## LEAVING AlphaCALC

To leave AlphaCALC, from the Main Menu, select the QUIT command.

When you leave AlphaCALC, you return to AMOS command level, or to the menu you started AlphaCALC from.

## AlphaCALC's DESIGN

The following diagram gives you a general idea of AlphaCALC's overall structure:




When you call up AlphaCALC, you're at the Main Menu. As the diagram shows, from the Main Menu you can go to your worksheet display or use a number of other commands which affect entire worksheets. These commands let you print, copy, or erase worksheets, see a list of worksheets, and so on.

At your worksheet display, you can enter new information or formulas, edit the data already in your worksheet, and use the function keys on your keyboard to perform editing functions, such as erasing or inserting rows and columns, and so on.

From your worksheet display, you also have access to commands which affect only the current worksheet. You can invoke these features directly from your worksheet using slash commands, or by going to the AlphaCALC Editing Menu and choosing the command you want. These commands let you print your worksheet, sort sections of it, search for particular entries, and much more.

# CHOOSING AlphaCALC COMMANDS

This section describes how to select any available AlphaCALC command, from the Main Menu, Editing Menu or your worksheet display.

## TO CHOOSE A MENU OPTION

You can choose any command on the Main Menu or Editing Menu in either of these ways:

- Type the command name and press [RETURN].
- Highlight your choice using [TAB] and [BACK TAB] or [↑] and [↓] and press [EXECUTE].

When typing the command name, you only have to type enough to distinguish it from other commands. For example, on the Main Menu, you can type just `Q` for **QUIT**, since no other command starts with Q. To choose **COPY**, type `CO`, since **CREATE** also begins with C.

### Command Type Ahead

For all commands where the first thing you enter on the command screen is a worksheet name—**CREATE**, **EDIT**, and so on—if you type the command name to select the command, you can include the name of the worksheet you want. For example, to **EDIT** the worksheet named MONTH, you can type this command at the Main Menu:

**ED MONTH** [RETURN]

## TO USE A COMMAND FROM YOUR WORKSHEET

You can invoke any Editing Menu command (except **HELP**) directly from your worksheet display by using a "slash" (/) command. This is merely an alternate way to use these features; slash commands and Editing Menu commands let you do exactly the same things, so use whichever method is easier for you.

One feature, inserting page breaks, is available only as a slash command, not from the Editing Menu.

To use a slash command, at the worksheet display:

- Type / and the letter(s) for the option you want.

If you type just a /, The Worksheet Command Menu displays the available options. To choose an option, type its first letter. If another window of options appears, again type the first letter of your choice. You don't need to press [RETURN] or [EXECUTE] when selecting a slash command.

When you use a slash command, the command screen window appears over your worksheet display, and you return to your worksheet when the command is finished.

# USING AlphaCALC SCREENS

This section describes the general procedures for using any screen in AlphaCALC; unless stated otherwise, the methods given here work on any menu or command screen.

## MOVING AMONG FIELDS

On screens with more than one field, use:

- [TAB] or [RETURN] to go to the next field. You can normally use [↓] also.
- [BACK TAB] to go to the previous field. [↑] also works in most cases.

## MAKING SELECTIONS

In a field where you select from a limited number of choices, such as the **DIRECTORY** command's format field, you can select the choice you want in either of these ways:

- Type its first letter.
- Press [NEXT WORD] or [PREV WORD] to cycle through the choices.

In fields with larger numbers of possible entries, such as the **FORMAT** command's format field, you need to type the entire entry.

## ENTERING A WORKSHEET NAME

In any worksheet name field, you can either type the name of the worksheet you want, or press [↑] or [↓] to see a list of worksheets. While the list is displayed, you can press:

- [NEXT SCREEN] to scroll the list. When you've reached the end of the list, it will "wrap around" to the beginning.
- [↑] or [↓] to highlight the name you want.
- [EXECUTE] to choose the highlighted worksheet.
- [MENU] to return to the worksheet name field without choosing a worksheet name. You can then type the name you want.

### Worksheet Name Format

AlphaCALC worksheet names consist of from one to six letters and/or numbers. These are all allowable worksheet names: TEST, X, WEEK1, 123456, ABCXYZ.

You cannot use more than six characters in a worksheet name, or any character other than a letter or number.

You can enter either upper or lower case letters for a worksheet name. AlphaCALC treats them the same.

### TO PERFORM THE COMMAND

When you've made the choices you want on a command screen, to perform the function of the command, press [EXECUTE]. You can press [EXECUTE] at any time; you do not have to move the cursor through all the fields on the screen.

From most command screens, this returns you to the menu you came from; with some commands, you stay at the command screen so you can make further entries.

### TO LEAVE A SCREEN WITHOUT PERFORMING THE COMMAND

To leave a command screen without having any of the entries or changes since you last pressed [EXECUTE] take effect, press [MENU].

From the screen for a Main Menu command, you return to the Main Menu; after an Editing Menu command, you return to the Editing Menu; after a slash command, you go to your worksheet display.

In the Worksheet Header, [MENU] has the same effect as [EXECUTE]—you go to your worksheet display and the current settings are saved.

## USING KEY FEATURES

You use many of AlphaCALC's worksheet editing features by pressing a key while at your worksheet display. AlphaCALC uses both the function keys, which are labeled with F and a number and found at either the top or left side of your keyboard, and the "special" keys, such as [EXECUTE], [NEXT SCREEN], and so on.

All references to function key features use the name of the AlphaCALC feature, rather than the "F number" of the key. Because different types of terminals have different numbers of function keys, a feature you use by pressing [F4] on one terminal may be [F14] on another. However, the AlphaCALC feature name is always the same.

Keyboard overlays showing the AlphaCALC feature name of each function key are available for all common Alpha Micro terminals and 10- and 12-function key PC keyboards.

If your terminal does not have a key we refer to—for example, not all keyboards have a [HELP] key—you can use the "universal key sequence" for the feature. The universal key sequence involves the [CTRL] and/or [ESCAPE] keys, and works on any type of terminal. In Part 2, the section for each key feature includes the universal key sequence for the feature. Appendix D lists all the universal key sequences.

*AlphaCALC Keyboard Command Quick Reference* lists the keys to use for any AlphaCALC feature on all Alpha Micro supported terminal types, including PCs, and the universal key sequence for each feature.

## HELP INFORMATION

All of AlphaCALC's screens include messages at the bottom telling you how to proceed. If you want additional information on any topic, you can view help information from anywhere in AlphaCALC. You have these choices:

- From the Main or Editing Menu, select the **HELP** command. You see a new menu, listing help topics relating to the commands on the menu. Select the one you want by typing its number or highlighting it, then press [EXECUTE].

- From anywhere in AlphaCALC, you can press [HELP]. What displays next depends on where you are:

    - From the Main or Editing Menu, you see information on using the menus.
    - From any command screen, you see information on that command.
    - From your worksheet display, you see a list of topics relating to your worksheet. Select the one you want.
    - If you're at one of the cell editing fields, you see information about making cell entries.

While you're looking at the help information, you can use [NEXT SCREEN] and [PREV SCREEN] to scroll through the help screens.

When you're through with the help information, press [MENU] to return to where you were.

## THE WORKSHEET DISPLAY

The worksheet display on your terminal screen is divided into the following sections:

- The worksheet name and description may appear on the top line. Only certain terminals support this feature.

- Most of the screen shows the cells of your worksheet—as many as will fit. The cursor highlights the current cell.

Each cell is identified by a unique cell ID consisting of its layer, column, and row. Cell IDs are discussed later in Part 1.

Borders at the top and left of your screen indicate which columns and rows are displayed. You can turn these borders on and off. The next section tells you how to display a different part of your worksheet.

- Beneath the worksheet are the cell editing fields. These lines describe the current cell—its formula, value, format, alignment and color.
- The bottom line of the display shows the current cell ID at the far left. The rest of the line is reserved for error and other messages and indicators showing what options—such as automatic cursor advance—you have selected.

## MOVING AROUND THE WORKSHEET

The average terminal screen can show approximately nineteen rows and eight columns of one layer of your worksheet. The largest possible worksheet can have 255 layers of 8191 rows by 255 columns each. Obviously, you need ways of moving the screen to see other parts of your worksheet.

You can move around the worksheet in any of the following ways.

| TO MOVE THE CURSOR | DO THIS |
|---|---|
| One cell in any direction: | Press [↑],[↓], [→] or [←] |
| To a specific cell: | Press [GOTO]. Type **`Cell ID`** [RETURN] [1] |
| To the beginning of the row: | Press [SHIFT] / [←] |
| To the end of the row: | Press [SHIFT] / [→] |
| To the beginning of next row: | Press [RETURN] |
| To the top left of the screen: | Press [SHIFT] / [↑] |
| To the bottom right of the screen: | Press [SHIFT] / [↓] |
| To cell A1 of current layer: | Press [HOME] |
| To the bottom of this layer: | Press [SHIFT] / [HOME] |
| To the next cell with data:[2] | Press [TAB] or [NEXT WORD] |
| To the previous cell with data:[2] | Press [BACK TAB] or [PREV WORD] |

1 See the section on [GOTO CELL] in Part 2.

2 The direction the "next" cell is in is determined by the last direction you set using [AUTO-ADVANCE]. These features skip protected cells.

All of the above procedures change the part of the worksheet shown on your screen as necessary to show the new location of the cursor.

Also, to see how to have the cursor automatically move to the next cell when you enter a cell formula, see the section on [AUTO-ADVANCE] in Part 2.

You can change the displayed section of your worksheet in any of these ways:

| TO MOVE THE DISPLAY | DO THIS |
|---|---|
| One screen down: | Press [NEXT SCREEN] |
| One screen up: | Press [PREV SCREEN] |
| One screen right: | Press [SCREEN RIGHT] |
| One screen left: | Press [SCREEN LEFT] |
| To the next layer: | Press [SHIFT] / [NEXT SCREEN] |
| To the previous layer: | Press [SHIFT] / [PREV SCREEN] |
| Center the current row: | Press [CENTER SCREEN] |

## ABOUT CELL IDs

Every cell in an AlphaCALC worksheet is identified by a unique cell ID. You often need to enter these IDs, either because you want to go to the cell, as part of a cell formula, or while using a command such as **DUPLICATE**. The cell ID is in the format:

Layer|ColumnRow

Where:

- Layer is the layer letter(s). There are 255 possible layers, so layer letters are in the range A - Z, AA - AZ, BA - BZ, and so on to IA - IU.

  If you leave out the layer letter, AlphaCALC assumes you mean the current layer. If you include the layer letter, you must follow it with the | (split vertical bar) character.

You can change the character used after the layer letter either in the AlphaCALC initialization file or by using the **MODE** command.

- Column is the column letter(s). Possible column letters are the same as the layer letters.

- Row is the row number, in the range 1 to 8191.

You must include both the column letter and the row number. These are all legal cell IDs:

```
A|A1        C|AB12      F17         B612        IU|IU8191
```

### CELL RANGES

Many AlphaCALC commands and functions require you to enter a cell range. A cell range is a rectangular area of your worksheet, and you specify the range you want by entering the top left and bottom right cells in the area. The format for a cell range is:

```
ID:ID
```

For example, A1:F12 specifies the range with cell A1 at the top left and cell F12 at the bottom right. If you don't include the layer letter, as in this example, AlphaCALC assumes you mean the current layer.

You can enter a range covering two or more layers—for example A|D3:C|G6. This means the cells in the range D3:G6, on layers A, B and C (sixteen cells on each layer, a total of 48).

#### Ranges on Command Screens

When you are asked for a cell range on many command screens, you have these choices for your entry:

- A cell range, as described above.
- A layer letter followed by |, to designate an entire layer. For example, B|.
- A column letter to mean an entire column on the current layer only.
- A row number to mean an entire row on the current layer only.

Not all commands which request a cell range will accept a layer, row or column entry.

## ENTERING INFORMATION IN A CELL

To enter information—a number, text, a math formula, etc.—in a worksheet cell:

1. Move the cursor to the cell you want.

2. Type the information. It is displayed in the Formula field in the cell editing area.

   While you are making an entry, you are in editing mode. The cursor disappears from the worksheet cell and another cursor appears in the cell editing fields. You can't move to another cell or leave your worksheet display while you're in editing mode.

   Your entry can be up to 254 characters long. For the types of information and formulas you can enter, see "Types of Cell Formulas," below.

   If you decide you don't want to make any entry in this cell after all, press [MENU] or [CANCEL] to cancel your entry.

3. If you don't want to change any of the cell format fields, press [EXECUTE].

4. If you do want to change the cell format, alignment, or color, press [TAB], [BACKTAB], [↑], [↓] or [RETURN] to move the cursor to the field(s) you want to change, and make your choices. When you've set all the fields you want, press [EXECUTE].

After you type your entry, AlphaCALC may add a character to the beginning of it, indicating the type of formula. See "Forcing a Cell Formula Type," below.

For more information on cell formats, see "Cell Formats," later in Part 1. For ways to speed up formula entry, see "Auto Advance" and "Speed Mode" in Part 2.

## TO CORRECT ERRORS IN A FORMULA

To correct any typing mistake you make while entering a cell formula, you can do the following:

| TO DO THIS | PRESS |
|---|---|
| To move cursor right: | Press [→] |
| To move cursor right, erasing characters: | Press [SPACE] |
| To move cursor left: | Press [←] |
| To move cursor left, erasing characters: | Press [RUBOUT] |
| To erase characters: | Press [DEL CHAR] |
| To insert spaces: | Press [INS CHAR] |
| To erase from cursor to end of line: | Press [CTRL] / [Y] |
| To enter auto insert* mode: | Press [CTRL] / [Q] |
| To cancel entry and leave editing mode: | Press [CANCEL] or [MENU] |

* See the Auto Insert Mode section in Part 2.

## CHANGING AN EXISTING CELL

To change an entry in a cell, do this:

1. Move the cursor to the cell you want to change.

2. Press [EDIT CELL]. This puts you in editing mode. Move the cursor to the part of the entry you want to change and make your changes, as described above.

   If you prefer, instead of pressing [EDIT CELL], you can just start typing your new formula. What you type will replace the previous entry, character for character.

3. If you don't want to change any of the cell format fields, press [EXECUTE].

To cancel your changes and return the cell to how it was before you edited it, press [MENU] or [CANCEL] instead of [EXECUTE].

4. To change the cell format, alignment, or color, press [TAB], [BACKTAB], [↑], [↓] or [RETURN] to move the cursor to the field(s) you want to change, and make your choices. When you've set all the fields you want, press [EXECUTE].

## CELL FORMULAS AND CELL VALUES

At various times, the AlphaCALC software and documentation refer to cell formulas and cell values. These terms have specific meanings within AlphaCALC:

- The cell formula is whatever you type in the formula field for the cell.
- The cell value is whatever is displayed in the value field in the cell editing area. When a cell formula is a constant (see "Types of Cell Formulas," below), the cell value and cell formula are the same, and nothing is displayed in the value field.

The worksheet cell displays the *formatted* cell value. This is normally the same or very similar to the actual cell value (for example, it may be changed to a percentage or have a dollar sign in front of it), but in some cases, such as formatted dates, it may be very different.

For example, if you type 77 in cell A1, 77 is both the cell formula and the cell value. If you then type A1 in cell B3, the cell formula is A1 (what you typed), but the value is 77 (the value of A1, which it refers to).

### TYPES OF CELL VALUES

There are three kinds of cell values:

- Numeric: for example 1, .0054, or -8976.4
- Logical: true or false.
- String: for example "John Smith," or "Quarterly Profits." In the Value field, strings are always shown in quotes.

The cell value is sometimes referred to as the cell contents.

## TYPES OF CELL FORMULAS

Cell formulas can be divided into two main types:

- Constants: numbers or text—anything where what is displayed in the cell is the same as what you typed in the cell (except for formatting differences, as discussed above). When you enter a constant, nothing is displayed in the value field.
- Expressions: all other formulas.

Expressions can be further divided based on the kind of value they give when they're evaluated—numeric, logical, or string.

An expression consists of a combination of the following:

- Constants—numbers or text strings.
- Cell references.
- Arithmetic operators, such as + or -.
- Logical operators, such as = or <.
- The string operator, &.
- AlphaCALC functions, such as SUM.
- Glossary items.

For example, these are all legal expressions:

```
A7 + 19.4
6 < B8
SUM(B|G5:C|I8)
IF(K13<K12,AVG(K12:K15),"Incorrect")
```

It is the power and flexibility of the expressions you can enter that makes AlphaCALC a useful calculating tool.

The following sections describe the various components of expressions.

## CELL REFERENCES

A cell reference lets you use the value of one cell as all or part of the value in another cell. To use a cell reference, type the ID of the cell you want to refer to. For example, if cell B5's value is 42, and you use B5 as the cell formula in cell D12, then D12's value is also 42. If B5's value changes, when you recompute your worksheet, so does the value of D12.

There are two types of cell references in AlphaCALC, relative and absolute. The only difference between them is what happens when you move or copy the cell they're in.

A relative cell reference always refers to the cell in the same relative position to the cell it's in. In the example above, a relative reference to cell B5 in cell D12 always refers to the cell two columns left and seven rows up from itself. If you copy cell D12 to cell J20, the reference changes to cell H13. This adjustment takes place whenever the cell containing the reference is moved by any AlphaCALC feature.

An absolute cell reference, however, never changes because the cell containing it moves. It always refers to the same cell. To mark a cell reference as absolute, enter a dollar sign ($) before each part of the cell ID. For example: $B$5 or $F|$G$45. Neither of these references will change, no matter how you move or copy the cell containing them.

You can have a mixed relative and absolute reference by placing the $ before only part of the cell ID. For example, $F6 will always refer to column F, but will adjust the row reference when moved, while K$56 will change its column reference, but not the row.

You can use relative, absolute or mixed cell references both in simple cell references, and when entering cell ranges or lists for AlphaCALC functions (see Part 3).

When you leave the layer ID out of a cell reference, it refers to a cell on the current layer. You need to include the layer ID only to refer to cells on other layers.

### Formula Adjustment

As described above, when you move or copy a cell containing a relative cell reference, the cell reference is adjusted so it refers to the cell in the same position relative to its new location, while an absolute cell reference doesn't change when you move the cell containing it. However, there is a circumstance when both relative and absolute cell references are adjusted.

Say you have a cell reference in cell A5, referring to cell B10. If you insert a row into your worksheet between rows 8 and 9, the value in cell B10 moves to cell B11. Where should cell A5 now refer to?

AlphaCALC assumes that when you referred to cell B10, you wanted to refer to the value in that cell, not necessarily that cell location. So, when you move a value by inserting or deleting a row or column, all references to the value are adjusted to account for its new location (in our example, B11). This is true for both relative and absolute cell references.

References to cells in a row or column you delete are not adjusted. This may cause problems in some worksheets, so be careful when deleting cells which are referred to in other formulas.

This doesn't contradict what we said above about absolute references—they are never adjusted because you move or copy the cell containing the reference. They are, however, adjusted to refer to the same value if you move the cell containing the value they refer to.

This "moving the value" cell reference adjustment occurs only when you insert or delete rows or columns. It does ***not*** take place when you copy cells using BLOCK COPY or **DUPLICATE.**

## OPERATORS

Operators are symbols you place between two values or expressions to perform an operation or comparison on them. The most familiar operators are the simple arithmetic ones, like the plus and minus signs (+ and -). AlphaCALC uses these operators, and many more. There are three types of operators: arithmetic, logical, and string.

### Arithmetic Operators

Arithmetic operators go between two numeric values and combine them in some way. The operators AlphaCALC uses are:

| | |
|---|---|
| + | Addition |
| - | Subtraction |
| * | Multiplication |
| / | Division |
| ^ or ** | Exponentiation |
| MOD | Modulo. Gives the remainder when the first number is divided by the second. |

### Logical Operators

Logical operators go between two numeric expressions (simple operators), or two logical expressions (compound operators). They compare the expressions in some way, and their value is either TRUE or FALSE (except for the MIN and MAX operators).

The simple operators are:

| | |
|---|---|
| = | Is equal to |
| <> | Is not equal to |
| > | Is greater than |
| < | Is less than |
| <= or =< | Is less than or equal to |
| >= or => | Is greater than or equal to |
| MAX* | Which value is larger? |
| MIN* | Which value is smaller? |

* MAX and MIN display the selected value, not TRUE or FALSE.

If you use the second version of any of these, AlphaCALC changes the display to the first version.

The compound operators are:

| | |
|---|---|
| AND | Are both expressions true? |
| EQV | Are the expressions the same (both true or both false)? |
| IMP | Does the truth of the first expression imply the truth of the second?* |
| NOT | This works with only one logical expression. It reverses its value. |
| OR | Are either or both expressions true? |
| XOR | Is one and only one of the expressions true? |

* IMP is true if both expressions are true, or if the first expression is false. If the first expression is true and the second is false, then IMP is false.

### String Operator

There is one AlphaCALC operator which lets you work with strings. The ampersand (&) operator joins (concatenates) two strings. For example "Alpha"&"CALC." = "AlphaCALC."

### Operator Precedence

When evaluating an expression, AlphaCALC uses this order of operator precedence:

| PRECEDENCE | OPERATORS |
|---|---|
| 5 | ^ |
| 4 | *, / |
| 3 | +, - |
| 2 | <, >, =, <>, <=, >= |
| 1 | &, MOD, MIN, MAX, AND, EQV, IMP, OR, XOR |

Operators with a higher precedence are performed first; operators with the same precedence are performed from left to right in the expression. For example:

```
34+5*7<4^2/2*8
```

The exponentiation (4^2=16) is done first. Then, multiplication and division are performed from left to right (5*7=35, 16/2=8, 8*8=64). Then, the addition is done (34+35=69). Last, the logical comparison is checked (69<64= false).

### Using Parentheses to Change Precedence

If you want to change the order in which operations are performed in an expression, put the parts of the expression you want calculated first inside parentheses. You can use as many sets of parentheses as you want. Operations inside the most parentheses are performed first. Within parentheses, the order listed above is used.

For example, if we added parentheses to the expression above:

```
(34+5)*7<4^2/(2*8)
```

Now, the operations in parentheses are done first (34+5=39 and 2*8=16). Then the exponentiation (4^2=16), then the multiplication and division outside the parentheses (39*7=273 and 16/16=1). Finally, the logical comparison is 273<1, and the answer is false.

## FUNCTIONS

A function is a shorthand way to perform a mathematical or other operation in a cell formula. You enter the name of the function, and AlphaCALC understands it to mean "Perform such and such operation." For example, SUM tells AlphaCALC to add a group of numbers.

There are almost a hundred functions available in AlphaCALC. They are discussed in Part 3.

## GLOSSARY ITEMS

Glossary items let you take certain information stored by your computer and use it in your worksheet. Most of the glossary items are technical in nature; see the *AlphaCALC Programmer's Information Manual* for a complete list of them.

## FORCING A CELL FORMULA TYPE

Normally, you type the information you want in a cell, and AlphaCALC automatically figures out what type of cell formula it is. If you type `167.5`, AlphaCALC treats it as a number; `Quarterly Profits` is text, and so on.

However, you may sometimes want to tell AlphaCALC your entry is a particular type of formula. There are two common reasons for this:

- When you want an entry which could be a numeric formula to be treated as text.
- If you make a format error in a complicated formula, AlphaCALC may treat the formula as text, since it can't properly evaluate it. In this case, you want to force AlphaCALC to consider the entry as an expression, since it may help you find and correct the mistake.

You can force AlphaCALC to consider your entry as a specific type of formula by beginning it with one of these characters:

| | |
|---|---|
| = | Expression |
| " | Text - right justified |
| ' | Text - left justified |
| ^ | Text - centered |
| \ | Text - duplicating |

If you don't include one of these characters, AlphaCALC evaluates your entry, then adds the appropriate character at the beginning of the formula field.

## CELL FORMATS

The cell format determines how the cell value displays in the cell—it does not affect the actual value, only its appearance.

There are two components to cell format:

- The format: how the value displays—dollars and cents, scientific notation and so on—and the number of decimal places it's rounded to.
- The alignment: where the value appears in the cell—right or left justified or centered.

### SETTING CELL FORMAT

To set the cell format while you're entering a cell formula:

1. After typing the formula, press [TAB], [↓], or [RETURN] to go to the Format field.
2. Type the code for the format you want (see the next section).

3. If you want to set the cell alignment, press [TAB], [↓] or [RETURN].

4. Type the code for the cell alignment you want.

5. Press [EXECUTE].

You can set the cell format for a cell range or your entire worksheet using the **FORMAT** command. See the section on **FORMAT** in Part 2.

## TYPES OF CELL FORMATS

The standard cell formats available for numeric values in AlphaCALC are:

| CODE | DESCRIPTION |
|---|---|
| GEN | General. No rounding, no special formatting. |
| I | Integer. |
| IC | Integer, with commas. |
| N1 | Rounded to one decimal place. |
| N2 | Rounded to two decimal places. |
| N2C | Rounded to two decimal places, with commas. |
| N3 | Rounded to three decimal places. |
| N4 | Rounded to four decimal places. |
| N5 | Rounded to five decimal places. |
| $C | Preceding dollar sign, with commas. |
| $2 | Dollar and cents, with dollar sign. |
| $2C | Dollars and cents, with dollar sign, with commas. |
| % | Percentage. |
| %1 | Percentage, rounded to one decimal place. |
| %2 | Percentage, rounded to two decimal places. |
| %3 | Percentage, rounded to three decimal places. |
| %4 | Percentage, rounded to four decimal places. |
| %5 | Percentage, rounded to five decimal places. |
| E2 | Scientific notation, rounded to two decimal places. |

Notes:

- All currency formats put negative values in parentheses.
- The "dollar sign" is the currency symbol for the currently selected language.
- "With commas" means the thousands separator symbol for the currently defined language is placed every three digits.

In addition to these, you can define other cell formats for numbers in your AlphaCALC initialization file. See the *AlphaCALC Installation and Administrator's Guide*.

These formats are available for date and time values:

| CODE | DESCRIPTION |
|---|---|
| D1 | Month name DD, 19YY |
| D2 | DD Month name, 19YY |
| D3 | DD-Mon-YY |
| D4 | MM/DD/YY |
| D5 | DD/MM/YY |
| T1 | HH:MM AM/PM |
| T2 | HH:MM (24 hour) |
| T3 | HH:MM:SS AM/PM |
| T4 | HH:MM:SS (24 hour) |

These formats work only on ***serial*** date and time values, which are returned by several of the functions described in Part 3. Specifically, they do not properly display the $TM and $TD glossary items, which return the date and time in a different format. See *AlphaCALC Programmer's Information* for more information on glossary items.

### TYPES OF CELL ALIGNMENT

There are five alignments available in AlphaCALC:

| CODE | DESCRIPTION |
|---|---|
| G | General: right justified numbers and centered text |
| C | Centered |
| R | Right justified |
| L | Left justified |
| D | Duplicating |

Duplicating alignment deserves special mention. It repeats the cell contents until they fill the entire cell. It is generally used with single character text entries—especially dashes, equal signs, asterisks and so on—to create total lines and other separators to make the worksheet more readable.

You can set the alignment of a text entry in the formula field. See "Forcing a Cell Formula Type," above.

## HOW AlphaCALC DISPLAYS INFORMATION

The previous sections described how you can control the appearance of values in your worksheet. But, what if your requested format and the data in your worksheet don't match? For example, what if you enter a seven digit number in a column that's only five characters wide?

When an entry is too large to fit in a cell, AlphaCALC handles text and numbers differently.

For text, AlphaCALC displays as many characters as it can. The display continues until it reaches another cell with data in it. For example, if column A is nine characters wide, and you enter "Column heading" (fourteen characters) in it, the last five letters will display in column B. If the cell in column B has data in it, then AlphaCALC will truncate the entry in column A, and you'll see `Column he`.

For numbers, AlphaCALC will do one or both of two things:

- It will put the number in scientific notation, as in the E2 format discussed above. For example, if an entry of 1,000,000 is too long for its cell, AlphaCALC will display `1E6`.

- If the number has too many significant digits to fit, AlphaCALC will display as many characters as it can, and put an "overflow character" at the beginning of the cell to let you know not all the value is shown. The default overflow character is #, but you can change it in your initialization file or using the **MODE** command.

  For example, if you enter 1234567.89 in a seven character column, AlphaCALC displays `#123456`.

When you perform operations requiring extreme precision—generally, more than eight decimal places—you may see slight rounding errors. This is a limitation of the "floating point" method AMOS uses to store numbers. To avoid these discrepancies, you can set the cell format as described above or use the ROUND function described in Part Three.

Also due to limitations of floating point numbers, the largest number AlphaCALC can handle is approximately 1.7014E38. Attempting to use numbers larger than this will cause an error message.

## COLOR IN AlphaCALC

If you have a color terminal and/or printer, you can set the colors for various parts of your worksheet display:

- You can set the row and column border colors in your AlphaCALC initialization file or the **MODE** command.

- You can set the color for a single cell when you enter the cell formula.

- You can set the color for a section of your worksheet using the **FORMAT** command.

- You can set the color for each data range on a graph using [GRAPH SETUP]

In all of these cases except graphs, you can set both the foreground and background colors.

AlphaCALC allows you to choose from eight colors, numbered zero through seven. While your screen lists a specific color for each number (0 = black, 1 = white, and so on), what you see may not match this list. The colors listed are the "standard" colors for each number. However, most color terminals let you choose display colors from a palette. Since two terminals on a system may display differently for the same color settings, there is no way for AlphaCALC to accurately list your color choices. You'll just have to try the various color numbers and see how they display on your screen.

Also, if you have a color printer, the printed colors may not match your display colors. Specifically, black and white are normally reversed. Again, try printing all the color selections to see what you get on your printer.

You can also set colors for other parts of the screen display by modifying the screen files. See the *AlphaCALC Installation and Administrator's Guide.*

# PART TWO
# COMMANDS AND KEYS

This section lists all of the commands and function key features you can use in AlphaCALC. The commands and keys are listed alphabetically. If you aren't sure which feature you need, check the index or scan the list of one line descriptions of each feature in Appendix A.

## AUTO ADVANCE

AlphaCALC's automatic advance feature speeds up formula entry by moving the cursor to the next cell when you finish a formula entry.

To turn on auto advance:

1. Press [AUTO-ADVANCE] or [CTRL] / [\].

2. Choose the direction you want the cursor to go. You choose a direction by typing its first letter. Since AlphaCALC worksheets are three dimensional, you have these choices:

   - Right or Left.
   - Up or Down.
   - In or Out.

   A complete direction set is three letters, one from each of these pairs. For example, the default direction is RDI. This moves the cursor to the right, then, at the end of the row, down to the next row, and, when this layer is finished, in to the next layer. Another common choice, DRI, moves the cursor down this column, then right to the next column, then in to the next layer.

To turn off automatic advance, press [AUTO-ADVANCE] again.

Automatic advance follows these rules:

- The cursor always skips protected cells.
- When you press [EXECUTE] after entering a formula in an empty cell, the cursor goes to the next empty cell.
- After you edit an existing cell, the cursor goes to the next cell with data. If you edit the last cell in your worksheet, the cursor doesn't advance.

The automatic advance direction also sets the direction for the [TAB] and [BACKTAB] keys when moving around your worksheet, and determines the order the **SEARCH** command looks through your worksheet in.

## AUTO INSERT MODE

When you are in auto insert mode, characters you type in the formula field do not write over existing characters in the cell formula. Instead, existing characters are moved to the right to make room for your entry. This can be very useful when you leave one or more characters out of a cell formula, and don't want to have to retype the whole thing.

To turn on auto insert mode, while the cursor is in the cell formula field, press [CTRL] / [Q].

Auto insert mode affects only the formula field. It stays on as long as you are in your worksheet, or until you turn it off by pressing [CTRL] / [Q] again.

## BLOCK CLEAR

The [BLOCK CLEAR] function key clears any existing marked cell range. You should always clear block marks when you are done working with the marked range.

To clear block marks, at your worksheet display, press [BLOCK CLEAR] or [ESCAPE] [Q].

## BLOCK COPY

The [BLOCK COPY] key lets you copy a marked block of cells to another location in your worksheet. To copy a block of cells:

1. Move the cursor to the top left cell of the range you want to copy.
2. Press [BLOCK MARK].
3. Move the cursor to the bottom right cell of the range.
4. Press [BLOCK MARK].
5. Move the cursor to the top left cell of the destination range.
6. Press [BLOCK COPY] or [ESCAPE] [C].

Keep these things in mind when copying a block of cells:

- Any cells in the range you're copying to are copied over, so be careful not to erase any parts of your worksheet you want to keep.
- All relative cell references are adjusted to account for the new locations of the cells. Make sure any cell references you don't want to adjust are absolute references.
- The copied cells are not automatically recalculated to account for their new positions. You may want to use [RECOMPUTE] after the copy.

Copying a block of cells does not change the original cells in any way.

If you want to make more than one copy of a cell range, you can repeat steps 5 and 6, or use the **DUPLICATE** command.

## BLOCK ERASE

The [BLOCK ERASE] key lets you erase all unprotected cells in a cell range. To use it:

Be sure to clear any existing block marks before marking the range you want to erase.

1. Move the cursor to the top left cell of the range you want to erase.
2. Press [BLOCK MARK].
3. Move the cursor to the bottom right cell of the range.
4. Press [BLOCK MARK].
5. Press [BLOCK ERASE] or [ESCAPE] [D].

If any cells in the range are protected, they are not erased.

## BLOCK MARK

Marking a block of cells (also called a cell range) is one of the most common AlphaCALC operations. Once you've marked a block, you can copy it, erase it, sort it, change the cell format, and many other things.

A marked block of cells is always rectangular—if it includes multiple layers, it is a rectangular section of each layer. You cannot mark any other shape.

To mark a block of cells:

1. Move the cursor to the top left cell of the range you want to mark.

2. Press [BLOCK MARK] or [CTRL] / [P].

3. Move the cursor to the bottom right cell of the range.

4. Press [BLOCK MARK].

To mark a three dimensional block—one covering two or more layers—mark the front top left cell and the back bottom right cell.

When a block is marked, the starting and ending cell IDs are shown on your terminal's bottom status line.

You can expand a marked block by moving the cursor to where you want the new boundary and pressing [BLOCK MARK]. To decrease a marked block, you must use [BLOCK CLEAR], then mark the new block you want.

When you're done working with a marked block, be sure to use [BLOCK CLEAR] to clear the block marks. Otherwise, later operations may affect only the marked block when you want them to affect the entire worksheet.

## BORDER COLUMN

The [BORDER COLUMN] key turns the column border at the top of your worksheet display on and off. You may want to turn off the border display so you can see an extra row of your worksheet on the display.

To turn off the border, press [BORDER COLUMN] or [ESCAPE] [B]; to turn the border back on, press [BORDER COLUMN] again.

This key affects only your worksheet display, not printing your worksheet.

## BORDER ROW

The [BORDER ROW] key turns the row border at the left of your worksheet display off and on. Turning the border off lets you see approximately 5 characters more of each row on your screen. This is normally about half a column.

To turn off the border, press [BORDER ROW] or [ESCAPE] [b]; to turn the border back on, press [BORDER ROW] again.

This key affects only your worksheet display, not printing your worksheet.

## COPY

The **COPY** command lets you make an exact duplicate of an existing worksheet. The new worksheet is just like one created with the **CREATE** command—you can edit it, print it, and so on.

To use **COPY**:

1. Select **COPY** from the Main Menu.

2. Enter the name of the worksheet you want to copy. You can type the name, or press [↑] or [↓] to see a list of worksheets and select the one you want from the list.

   You can copy a worksheet from another account by including the device and account number with the worksheet name. For example: DSK2:BALANC[37,7].

   If the worksheet is password protected, you will be asked to enter the password. The new worksheet will be protected by the same password.

3. Type the name you want to give the new copy of the worksheet.

4. Press [EXECUTE].

You can also copy a worksheet using the prototype worksheet field in the **CREATE** command. This may be easier if you want to edit the copy immediately.

---

## CREATE

The **CREATE** command lets you make a new AlphaCALC worksheet. To **CREATE** a worksheet:

1. From the Main Menu, select **CREATE**.

2. Type the name you want to give the worksheet. A worksheet name consists of from one to six letters and/or numbers.

   If the worksheet name you enter already exists, you can press [NEXT SCREEN] to edit it.

3. The prototype worksheet is copied into the worksheet you create. The default prototype, DEFWKS, contains only standard formatting information, and creates a blank worksheet.

   If you want a different prototype, type its name, or press [↑] or [↓] to see a list of worksheets and select the one you want from the list.

   You can use a worksheet from another account as a prototype by including the device and account number with the worksheet name. For example: DSK2:BALANC[37,7].

4. Press EXECUTE.

The standard DEFWKS worksheet is in the ALC: account. If you want a different default worksheet, you can create a DEFWKS worksheet in this account, and it will be used as the default for this account. You can also modify the DEFWKS worksheet in ALC:, but be careful when doing so, since it may affect all AlphaCALC users on your computer.

When looking for the DEFWKS prototype worksheet, AlphaCALC follows the same search path it uses when looking for the AlphaCALC initialization file, as described in the *AlphaCALC Installation and Administrator's Guide.*

From **CREATE**, you go to the Worksheet Header for the new worksheet. When you've made all the changes and additions you want to the header information, press EXECUTE to go to your worksheet display.

## CUT

The **CUT** command lets you copy a cell range from your worksheet into its own worksheet. It is often used together with **PASTE** to move cells from one worksheet to another. To use **CUT**:

1. Select **CUT** from the Editing Menu or use **/FC** from your worksheet.

2. If there is a marked cell range in your worksheet, that range is the default. To change the range shown, type the cell range you want.

3. If you don't want to put the cells in the worksheet named TEMP, type the worksheet name you want.

4. Press EXECUTE.

If the worksheet name you enter already exists, you are asked if you want to overwrite it. If you do, all the current contents of the worksheet are replaced by the cell range you're cutting.

The cell range you cut is placed in the new worksheet, starting at cell A|A1. Relative cell references are adjusted to account for the cells' new positions.

## DELETE COLUMN

The DELETE COLUMN key deletes the column the cursor is currently in, and moves all columns beyond it one column to the left to fill in the space.

To use DELETE COLUMN:

1. Move the cursor to any cell in the column you want to delete.

2. Press [DELETE COLUMN] or [ESCAPE] [CTRL] / [D].

Deleting a column affects only the current layer.

[DELETE COLUMN] deletes all cells in the column, even if they are protected.

If you want to erase a column without moving any other columns, use [ERASE COLUMN].

---

## DELETE ROW

The [DELETE ROW] key deletes the row the cursor is currently in, and moves all rows below it up one row to fill in the space.

To use [DELETE ROW]:

1. Move the cursor to any cell in the row you want to delete.

2. Press [DELETE ROW] or [CTRL] / [Z].

Deleting a row affects only the current layer.

[DELETE ROW] deletes all cells in the row, even if they are protected.

If you want to erase a row without moving any other rows, use [ERASE ROW].

---

## DIRECTORY

The worksheet directory is a list of the worksheets in your current account. The **DIRECTORY** command lets you view or print the directory, or a portion of it you select. To use **DIRECTORY**:

1. Select it from either the Main or Editing Menu, or use **/FD** from your worksheet.

2. You have these options:

   - The format you want. Formats are:

     - Standard: name, description, creation and revision dates, protection status, and version.
     - Name: name and description only.
     - History: same as standard, plus last date revised and printed, author, operator, department, number of times revised and printed.

In the standard and name formats, a worksheet's information flashes on its tickler date. You can then look at the Worksheet Header for that worksheet to see what action you need to take.

- Display on screen or print? If you choose to print, see the Printing Options section, below.

- Optional selection criteria. These fields let you limit the worksheets included in the directory display, based on the worksheet name, description, author, operator, and/or creation date. Only worksheets matching all your entries in these fields are displayed. You can use one, all, or any combination of these fields.

  In all the fields except creation date range, all worksheets containing your entry are matched. For example, if you enter LEE in the Author field, you can see worksheets authored by Lee Joseph or Mike Leeds, but not Nancy Lemon.

3. Once you've chosen the options you want, press [EXECUTE].

4. If you're viewing the directory, you now see the first 15 worksheets (except in the history format, where you see only three). Press any key to view the next part of the list. After you finish viewing, you return to the Display Worksheet Directory screen.

### PRINTING OPTIONS

When you press [EXECUTE] after selecting the print option, you see the Printer Selection Screen. On this screen, you select the printer you want, if you have more than one available, and set other printing options. The options are:

- Do you want to print the directory now, or save it as a disk file only? If you save it on the disk, it has the name DIRECT and the first available extension in the series .000, .001 ....
- Do you want the file to be erased after it prints?
- What form name should the file use? This will almost always be NORMAL.
- How many copies do you want to print?

When you've made all the selections you want, press [EXECUTE]. You then return to the Display Worksheet Directory screen.

## DUPLICATE

Like [BLOCK COPY], **DUPLICATE** lets you copy a range of cells to a new location. **DUPLICATE** is more powerful and flexible than [BLOCK COPY].

**DUPLICATE**'s biggest advantage is that you can make multiple copies of a cell range at once. The size of the destination range determines the number of copies. For example, if the source is B3:B4 (two cells), and the destination range is C5:C11 (six cells), **DUPLICATE** makes three copies of the cells.

**DUPLICATE** also gives you the option of copying cell values instead of formulas. When you do this, all cell formulas in the source range are evaluated, and their values are copied to the new range as constants. If you have a SUM function in the source with a value of 212, only the 212 is copied to the destination range.

You can also copy blank cells with **DUPLICATE**. This is useful when the destination range includes cells with data you no longer want.

Be careful not to copy over any parts of your worksheet you want to keep. All cells in the range you're copying to are copied over, including protected cells.

To use **DUPLICATE**:

1. If you want, mark the cell range you want to copy. This saves you from entering it on the **DUPLICATE** screen.
2. Select **DUPLICATE** from the Editing Menu or use **/RD** from your worksheet.
3. Type the source cell range—if you didn't mark it first.
4. The default destination range is the same size as the source range, and starts at the current cursor position. If you want a different destination, type it. Remember, it can be any size.
5. Select the other options you want, then press [EXECUTE].

**DUPLICATE** does not recalculate the values of the cells you copy. You may want to use [RECOMPUTE] after using **DUPLICATE**.

## EDIT

The **EDIT** command lets you display an existing worksheet so you can view or change it.

1. From the Main Menu, select **EDIT**.
2. Enter the name of the worksheet you want. You can either type the name, or press [↑] or [↓] to see a list of worksheets, then select the one you want from the list. Press [EXECUTE].

   If the worksheet name you enter doesn't exist, you can press [NEXT SCREEN] to create it.

3. If the worksheet has a password, type the password and press [EXECUTE].

## EDIT CELL

The [EDIT CELL] key puts you into editing mode, so you can enter a new formula or change an existing formula in the current cell.

You would normally use [EDIT CELL] to modify an existing cell formula:

1. With the cursor in the cell you want to change, press [EDIT CELL] or [ESCAPE] [E].
2. Type your changes to the cell formula.

In general, you don't need to use [EDIT CELL]. To enter a new cell formula or change an existing one, just start typing. But, in some cases, you may want to use [EDIT CELL] so you can modify a formula without retyping the first characters.

For more information on entering and editing cell formulas, see "Entering Information in a Cell" in Part 1.

---

## ERASE

The **ERASE** command lets you erase any worksheet from your account. To do so:

1. Select **ERASE** from the Main Menu.
2. Type the name of the worksheet you want to erase. Or, you can press [↑] or [↓] to see a list of worksheets, then select the one you want from the list. Press [EXECUTE].
3. If the worksheet has a password, type it and press [EXECUTE].
4. Type YES when you are asked to confirm you want to erase this worksheet. Press [EXECUTE].

---

## ERASE COLUMN

The [ERASE COLUMN] key erases all the unprotected cells in the current column of the current layer. To use it:

1. Move the cursor to any cell in the column you want to erase.
2. Press [ERASE COLUMN] or [ESCAPE] [CTRL] / [V].

All cells in the current column—except protected ones—are now blank. No other cells in the worksheet are affected.

To delete an entire column and move other columns over to fill in the space, use [DELETE COLUMN].

## ERASE ROW

The [ERASE ROW] key erases all the unprotected cells in the current row of the current layer. To use it:

1. Move the cursor to any cell in the row you want to erase.

2. Press [ERASE ROW] or [ESCAPE] [CTRL] / [Y].

All cells in the current row—except protected ones—are now blank. No other cells in the worksheet are affected.

To delete a row and move the following rows up to fill in the space, use [DELETE ROW].

---

## EXPORT

The **EXPORT** command is available from both the Main and Editing Menus. It lets you take all or part of a worksheet and create a disk file containing its data in a different format. You can use this to share information between AlphaCALC and other spreadsheet packages or your own software.

You can create four types of files:

- DIF: Data Interchange Format. This is a standard format used to transfer information between spreadsheets. It contains cell values.
- SYLK: This is a different industry standard interchange format. It can contain either values or formulas.
- Cell ID,Data: This format places each cell ID and its formula on a single line in an ASCII text file.
- Row Format: This turns each row of your worksheet into a line in an ASCII text file. It includes cell formulas, not values.

For format information on these types of files, see the *AlphaCALC Programmer's Information* manual.

To use **EXPORT**:

1. Select it from the Main Menu or Editing Menu or type **/FE** from your worksheet.

2. If you came from the Main Menu, enter the name of the worksheet you want to export data from. If it has a password, you will be asked to enter it.

3. Enter the range of cells you want to export. The default is either the entire worksheet or any marked range of cells.

4. Select the type of file you want to create.

5. Enter the name of the file. **EXPORT** defaults to these extensions for each format:

| | |
|---|---|
| DIF: | .DIF |
| SYLK: | .SLK |
| Cell ID, Formula: | .CID |
| Row Format: | .ROW |

6. If you're exporting a DIF file, select whether you want to place cells in the DIF file in order by row or by column.

7. If you're exporting a Row Format file, indicate whether to separate cells with commas or spaces.

8. If you're creating a SYLK file, choose whether to export cell formulas or cell values.

9. When all the entries are correct, press [EXECUTE].

## FINISH

The **FINISH** command stores your worksheet in permanent memory on the disk and returns you to the Main Menu. You should use it after first creating a worksheet and whenever you leave a worksheet after making any additions or changes.

You can store your worksheet on disk without returning to the Main Menu by using the **SAVE** command.

To use **FINISH**, select it from the Editing Menu or type **/FF** from your worksheet display.

## FORMAT

**FORMAT** lets you determine how data appears in your worksheet. You can set the cell format, alignment, and colors, and change the width of columns, using **FORMAT**. And you can do so for any part of your worksheet: a single cell; a row, column, or layer; a cell range; or the default values for the entire worksheet.

The different types of cell format you can choose—monetary, percentages, and so on—are covered in the "Cell Formats" section in Part 1.

You can choose from among five cell alignments:

- Left justified.
- Right justified.
- Centered.
- General. This centers text and right justifies other values.
- Duplicate. The contents repeat to fill the cell width.

These alignments are also discussed in the "Cell Formats" section.

**FORMAT** does not affect protected cells.

To use **FORMAT**:

1. Select it from the Editing Menu, or type **/RF** from your worksheet display.

2. Select the area you want to affect. If there is a marked block in your worksheet, it is entered as the default.

   You can enter a cell range, column letter or row number. To specify a layer, enter the layer number followed by |. To change the default settings for the entire worksheet, type DEFAULT.

   When you set the format for a cell range, it affects only currently existing cells.

   If you set the format for a row, column, or layer, it affects existing cells and also changes the default settings for that row, column, or layer.

   Changing a row or column affects only the current layer.

   When you change the default setting for the worksheet, it does not affect existing cells. It also does not affect rows, columns, or layers for which you've entered format settings.

   If you enter a row, column, or layer, the default settings that appear are the current settings for the row, column or layer. But, if you enter a cell range, you see the default settings for your worksheet, even if you've previously changed the settings for that range.

3. Change the format, alignment and color settings to the ones you want. If you selected a column, a layer, or the whole worksheet, you can also set the column width.

   You can "hide" a column on your display by setting its width to 0. The column still exists, and you can still edit any cell in it. But, it doesn't appear on your screen or when you print, and the cursor disappears when you move it to a cell in that column.

4. You can set all cells in the range which contain the ACUM function to 0 by changing the `Clear ACUM` field to yes. See the section on ACUM in Part 3.

5. Press [EXECUTE]. You can then enter another area to format, or press [MENU] to leave this screen.

## GLOBAL

The **GLOBAL** command searches your worksheet or a cell range for a group of characters, and replaces each occurrence with the characters you specify.

**GLOBAL** can search all of your cell entries, or just certain types. Your choices are:

- Constants: only text entries and numbers. This includes only entries where nothing is displayed in the value field.
- Expressions: math, logical, or string expressions—any cell where AlphaCALC evaluates your entry and displays the calculated value in the cell. **GLOBAL** searches what you entered in the cell, not its displayed value.
- Formulas: all cell formulas, both constants and expressions.

For example: you enter 4 in one cell, and 4+3 (which displays as 7) in another. If you use **GLOBAL** to replace 4 with 6, searching constants only, only the first cell is changed. If you search expressions, only the second cell is modified—it now equals 6+3, or 9.

Constants and expressions are explained in "Types of Cell Formulas" in Part 1.

To use **GLOBAL**:

1. Select it from the Editing Menu or use **/WG** from your worksheet display.
2. The default range shown is either your entire worksheet or the currently marked block of cells. Type the range you want if this is not correct.

**GLOBAL** will not change protected cells.

3. Type the string you want to look for, and the one you want to replace it with.
4. Choose constants, expressions, or formulas, as discussed above.
5. You can choose to be asked to confirm each replacement. If you want to do so, change the `Global Replace?` field to **N**. This causes **GLOBAL** to work just like the **REPLACE** command.
5. When you’ve made all your entries and choices, press [EXECUTE].

If you’re doing a **GLOBAL** replacement, that’s all there is to it. If you’ve chosen to be asked for confirmation, AlphaCALC displays each cell containing the search string and asks if you want to replace it.

## GOTO CELL

The [GOTO CELL] key moves the cursor to any cell in your worksheet. To use it:

1. Press [GOTO CELL] or [CTRL] / [G].
2. Type the cell ID you want to go to and press [EXECUTE].

If you leave the layer number off the cell ID, the cursor goes to the indicated cell in the current layer.

If you type just a layer letter followed by | (for example, C|), you go to the current row and column on the layer you specified.

---

## GRAPH DISPLAY

The [GRAPH DISPLAY] key displays or prints the currently defined graph for your worksheet. To use [GRAPH DISPLAY]:

1. Press [GRAPH DISPLAY] or [ESCAPE] [G].

   The first time you press [GRAPH DISPLAY] after calling up your worksheet, you go to the Graph Setup screen. Press [GRAPH DISPLAY] again to display your graph.

2. If you have more than one graphics device defined, choose the device you want.
3. If the graphics device you chose is a printer, you can choose to save the graph as a disk file (formatted for the printer you selected), or to print multiple copies.

   If you select your terminal as the graphics device, the other fields have no effect.

4. Press [EXECUTE].

The graph is now displayed on your screen or printed on the printer. If you chose disk output only, the graph is saved as a file with the same name as the worksheet and an extension of .000 (or .001 if .000 already exists, and so on).

If you are using AlphaCALC under MULTI or AlphaOFFICE, you cannot display a graph on your screen, due to the way these products control your screen display. You can still print graphs to a printer.

# GRAPH SETUP

The [GRAPH SETUP] key lets you create five types of graphs—line, clustered bar, stacked bar, area, and pie charts—based on up to eight sets of data from your worksheet. You can then display the graph on a graphics terminal or print it on a graphics-capable printer.

## TO CREATE A GRAPH

In general, the process of creating a graph goes like this:

- Press [GRAPH SETUP] or [ESCAPE] [g] and start defining the graph.
- While defining the graph, you can go back and forth to your worksheet display without losing any information.
- Press [GRAPH DISPLAY] to see what the graph looks like.
- If you want, modify the specifications, then view the graph again. You can repeat this as often as you like.
- When the graph definition is complete, press [EXECUTE] from the Graph Setup screen to save it.

As long as you're editing your worksheet, the graph definition information is in the computer's memory, and you can display or modify the graph. However, the definition is saved ***only*** when you press [EXECUTE] on the Graph Setup screen. If you leave your worksheet without saving the graph definition the graph information is lost. A graph is not saved when you save your worksheet!

The Graph Setup screen is divided into four sections, each of which contains multiple fields where you define the various aspects of the graph. The display looks like this:

```
+---------------------  AlphaCALC Graph Set-up Screen ------------------------+
!                                                                             !
!   Graph Title: 1991 Sales ___________________________________________________ !
!   Subtitle #1: By Region                                                    !
!   Subtitle #2: First Quarter                                                !
!      Footnote: All figures preliminary                                      !
!  Y-axis Label: In Thousands                                                 !
!  X-axis Label: Widgits and Gizmos Only                                      !
!                                                                             !
! Y-axis starting value: 0.00000              Y-axis Grid type: Dotted grid   !
!   Y-axis ending value: 20.00000             X-axis Grid type: No grid       !
!             Increment: 2.00000                                              !
!                                                                             !
!            Graph type: Clustered Bar              Text Color: 1             !
!           Legend type: Legend                Background Color: 0             !
!                                                                             !
!           Data Range #: 1             4  data points                        !
!            Range color: 2                                                   !
!        Range linestyle: 1                                                   !
!             Cell range: F5:F8                                               !
!                  Title: Southwest                                           !
!                                                                             !
[ Press GRAPH to display graph, EXECUTE to save or MENU to return to worksheet]
```

### The Titles Section

The upper portion of the screen contains six fields where you can enter the titles and axis labels for the graph.

### The Axis Definition Section

The next five fields define the X and Y axes for the graph. You can set the starting and ending values for the Y axis, and the increments to be shown on the graph. You can also choose whether to include a dotted grid for both the X and Y axis.

If you set all the Y axis fields to 0, AlphaCALC sets the Y axis range and increment based on the data you're graphing. AlphaCALC always sets the X axis starting and ending values and increment based on the data.

These fields have no effect when you're creating a pie chart.

### The Graph Type Section

You can choose from five types of graphs:

- Line
- Clustered Bar
- Stacked Bar
- Area
- Pie Chart

This section also lets you choose whether to have a legend showing the title, line type and color for each data range (see the next section). If you do not include a legend, there is more room on the X axis, so the graph may be easier to read if you're plotting a lot of data.

You can also set the foreground and background colors for the graph. The normal settings of 0 for background and 1 for foreground result in a white on black graph on your terminal screen and a black on white printed graph.

### The Data Range Section

You can enter up to eight data ranges. The cells in each range (actually, it can be a range and/or list) contain one set of the values you want to graph. For example, on a line chart, each data range is graphed as a separate line.

For a pie chart, only the first data range is graphed.

The maximum number of points for each range is set in the AlphaCALC initialization file. You can change this maximum temporarily using the **MODE** command.

For area and stacked bar charts, the maximum number of points per range is 40, regardless of the AlphaCALC initialization file or **MODE** command settings.

For each range, you can choose a color, line style (which affects only line graphs), and title. If you include a legend on the graph, as described above, this information appears at the right side of the graph.

### Saving the Graph

You can go back and forth between the Graph Setup Screen, the displayed graph (by using [GRAPH DISPLAY]), and your worksheet display as often as you want while you're creating the graph.

As mentioned above, the graph definition data is not saved when you save your worksheet. To save the graph, you must press [EXECUTE] from the Graph Setup screen. This saves the data in a file with the same name as your worksheet and the extension .GDF.

If you want to create another graph from this worksheet while saving this one, you can go to AMOS command level and rename the .GDF file for the graph, then return to AlphaCALC and create the new graph. However, you'll need to leave AlphaCALC, go to AMOS, and rename the .GDF files whenever you want to view or modify the other graph.

## HEADER

The **HEADER** command takes you to the Worksheet Header screen. The Worksheet Header contains information about and sets parameters for your worksheet. For example, the page format of your worksheet when you print it is determined by settings in the Worksheet Header.

To see the Worksheet Header, select **HEADER** from the Editing Menu or type **/WH** from your worksheet.

You also see the Worksheet Header when you first **CREATE** a worksheet. The original settings in the Worksheet Header are taken from the prototype worksheet you use during **CREATE**.

The following lists describes the fields you can change in the Worksheet Header, divided into two categories: general information and page format settings. In addition, several fields at the bottom of the screen show the history of this worksheet: when it was created, when it was last revised and printed, and so on.

### GENERAL INFORMATION FIELDS

All of these fields (except Protection) are optional. However, the information on the worksheet directory is taken from these fields. If you leave a field blank here, it is blank on all directory displays.

- **Description**: Text, up to 60 characters.
- **Author:** Text, up to 20 characters.
- **Operator:** Text, up to 20 characters. Your system user name is automatically entered here, but you can change it if you want.
- **Department:** Text, up to six characters.
- **Protection:** You can enter N for no protection or P to assign a password.
- **Password**: This field appears only if there's a P in Protection field. Type a password of up to 10 letters and numbers. The password does not appear on the screen as you type it.

  The password is required whenever anyone tries to access the worksheet using EDIT, PRINT, or any other command.

If you assign a password, be sure to remember it! From inside AlphaCALC, there is no way to access a password protected worksheet without the password.

- **Tickler Date**: Enter a date on which the worksheet needs action. In the DIRECTORY command, the worksheet will flash on its tickler date.

  You can enter a date in any of these formats: MM/DD/YY; a day of the month (1 through 31); a day of the week (either a three letter abbreviation or the full word); or a specific day of a specific week of the month, such as 1st Tuesday or 4th Friday.
- **Action**: This text is associated with the tickler date. It can be up to 20 characters.

## PAGE FORMAT FIELDS

The fields below have to do with the page format when you print your worksheet. For a discussion of how these fields affect the printed worksheet, see the PRINT command section.

- **Lines per Inch**: Number of printed lines per inch.
- **Page length in lines**: The length of the paper you're using, in lines. For 8.5" by 11" paper in portrait mode, this is normally 66.
- **Top Margin**: Number of blank lines to put at the top of each page.
- **Bottom Margin**: Number of blank lines to leave at the bottom of each page.
- **Character spacing**: Number of printed characters per inch.
- **Left Paper Offset**: The left margin, in characters.
- **Characters per Line:** Number of characters to print per line.

- **Title Range:** You can enter a cell range to print as the title on each page. If you don't include a layer, PRINT takes the cells from the layer currently being printed.
- **Lines after title**: The number of blank lines to print after the title range.
- **Footer Range**: You can enter a cell range to print as a footer on each page. If you don't include a layer, PRINT takes the cells from the layer currently being printed.
- **Lines before footer**: The number of blank lines to print before the footer range.
- **Print Column border**: Should the column border be printed? Choose Y or N.
- **Print Row border**: Should the row border be printed? Choose Y or N.
- **Default Form:** The name of the form to print this worksheet on. NORMAL is the most common. You can change this selection during PRINT.
- **Orientation**: You can print the worksheet normally on the page (portrait), or, if your printer supports it, horizontally, to get longer lines (landscape). If you choose landscape mode, be sure to change the other page size settings to match the new page length and width.

## HELP

The HELP command and [HELP] key let you view help information from anywhere in AlphaCALC. Refer to "Help Information" in Part 1 for more information.

## IMPORT

The IMPORT command lets you take data from a non-AlphaCALC file and create an AlphaCALC worksheet containing it (from the Main Menu) or put the imported data in an existing worksheet (from the Editing Menu).

IMPORT accepts these four file formats:

- DIF: Data Interchange Format. This is a standard format used to transfer information between spreadsheets.
- SYLK: This is a different industry standard interchange format.
- Cell ID,Data: This format has each cell ID and its formula on a single line in an ASCII text file.
- Row Format: Each line of an ASCII text file becomes a row in your worksheet. Individual cells are separated by either columns or spaces.

For complete format information on these types of files, see the *AlphaCALC Programmer's Information* manual.

To use **IMPORT**:

1. Select it from the Main or Editing Menu or type **/FI** from your worksheet display.

2. Indicate the format of the file you're importing.

3. Enter the name of the file. You can enter a full file specification in this field. **IMPORT** defaults to these extensions for each format:

| | |
|---|---|
| DIF: | .DIF |
| SYLK: | .SLK |
| Cell ID,Formula: | .CID |
| Row Format: | .ROW |

4. If you're using **IMPORT** from the Main Menu, type the name you want for the new worksheet, and the worksheet you want to use as a prototype for it. If the prototype worksheet has a password, you'll need to enter it. See **CREATE** for more information on prototype worksheets.

   If you're using **IMPORT** from the Editing Menu, enter the top left cell of the destination range. The size of the destination range is determined by the size of the file you're importing. Be careful not to have the imported data overwrite existing cells you want to keep.

5. In Editing Menu **IMPORT**, you can choose whether to have the imported data overwrite any protected cells in the destination range.

6. If you're importing a DIF file, select whether you want to leave rows and columns as is, or change the DIF rows to columns in the worksheet and vice-versa.

7. If you're importing a Row Format file, indicate whether cells are separated by commas or spaces.

8. When all the entries are correct, press [EXECUTE]

## INSERT COLUMN

The [INSERT COLUMN] key lets you insert a blank column at the current cursor location. To use [INSERT COLUMN]:

1. Move the cursor to any cell in the column you want to insert a column before. For example, to insert a column between C and D, move the cursor to column D.

2. Press [INSERT COLUMN] or [ESCAPE] [CTRL] / [F].

[INSERT COLUMN] affects only the current layer.

## INSERT ROW

The [INSERT ROW] key lets you insert a blank row at the current cursor location. To use [INSERT ROW]:

1. Move the cursor to any cell in the row you want to insert a row above. For example, to insert a row between rows 5 and 6, move the cursor to row 6.

2. Press [INSERT ROW], [LINE INSERT], or [CTRL] / [B].

[INSERT ROW] affects only the current layer.

---

## ITERATION OPTION

See "Recompute - with Iteration" later in Part 2.

---

## LOAN

The **LOAN** command lets you see or print loan amortization information. You can choose from two types of information:

- An amortization schedule, which shows complete amortization information for a specific loan.

- An amortization analysis, which compares payments for loans of equal length but varying amounts and interest rates.

To use **LOAN**:

1. Select it from the Main or Editing Menu or type **/L** from your worksheet.

2. Select an amortization schedule or an amortization analysis.

3. For an amortization schedule, enter:

   - The loan amount, up to ten billion dollars.
   - The annual interest rate, in percent, not as a decimal. For example, for 18%, enter **18**, not **.18**. It can't be over 99.999%.
   - The length of the loan in months.
   - The starting date of the loan, in the proper format for your language. In the U. S., this is MM/DD/YY.
   - To print the schedule, set the `Output to printer` field to yes.

   When you've made all your entries, press [EXECUTE].

If you're printing the schedule, continue with step 5. If you aren't printing, the first year's information appears on your screen. The display is described in the section "The Amortization Schedule," below.

4. For an amortization analysis, enter:

   - The base loan amount, up to ten billion dollars.
   - The base annual interest rate, in percent, not as a decimal. For example, for 18%, enter **18**, not **.18**. It can't be over 99.999%.
   - The length of the loan in months.
   - The amount difference you want. This determines the increment between columns of the analysis table. It cannot be more than 10% of the base loan amount.
   - The interest rate difference. This determines the difference between rows of the analysis. It cannot be more than 1% of the base interest rate.
   - To print the analysis, set the `Output to printer` field to yes.

   When all your entries are correct, press [EXECUTE].

   If you're printing the analysis, continue with step 5. If you're displaying it on your screen, go to the section on "The Amortization Analysis," below.

5. Select the printer you want, and set the other printing options. The options are:

   - Do you want to print the information now, or save it as a disk file only? If you save it on the disk, it has the name LOAN and the first available extension in the series .000, .001, and so on.
   - Do you want the file to be erased after it prints?
   - What form name should the file use? This will almost always be NORMAL.
   - How many copies do you want to print?

   When you've made all the selections you want, press [EXECUTE]. The printed results are described below.

## THE AMORTIZATION SCHEDULE

The amortization schedule display shows one calendar year of the loan on your screen at a time. The top part of the screen displays the general loan information you entered, the monthly and yearly payment amounts, and the total interest payments for the year shown. The bottom half lists each payment, including: payment number, date, remaining balance, the amount of the payment applied to interest and the principal, and the total interest paid so far.

To see successive years, press any key.

The printed schedule is very similar to the screen display, but the interest due in each year is shown immediately after the last payment of the year. The general loan information prints at the top of each page.

### THE AMORTIZATION ANALYSIS

The top of the amortization analysis display shows the information you entered. Below it is a table showing the monthly payments for loans of varying amounts and interest rates.

The first column is for loans of the amount you entered. As you move to the right, the amount increases by the increment you chose. The top of the column shows the loan amount.

The top row shows payments for loans at the base interest rate. As you go down, the interest rate increases by the rate difference you entered. The interest rates are shown at the beginning of each row.

The monthly payment for a loan of a particular amount and interest rate is shown at the intersection of the column for the amount and the row for the interest rate.

You can use the arrow keys to "move" around the analysis like you move around the worksheet. They scroll one screen at a time.

The printed analysis looks just like the screen display. However, you only see as many rows and columns as fit on standard 8 1/2" by 11" paper, and no amounts or interest rates lower than the base entries are included.

## LOCK COLUMN

[LOCK COLUMN] lets you lock one or more columns on your worksheet display. The column(s) stay on your screen when you move to another part of your worksheet. This lets you keep report headings, etc., visible no matter what part of your worksheet you're working on.

To use [LOCK COLUMN]:

1. Move the cursor to any cell in the right most column you want to remain on your screen.

2. Press [LOCK COLUMN] or [ESCAPE] [L]. The border of the column the cursor is on, and all columns to the left of it, change to dim display.

3. You can move around your worksheet and edit cells, as usual. When you want to unlock the column, press [LOCK COLUMN] again.

If you try to move the cursor into a locked column, the screen scrolls so the column displays twice. You can change formulas in the "unlocked" copy of the column, but we suggest you unlock a column before editing any of its cells.

## LOCK ROW

[LOCK ROW] lets you lock one or more rows on your worksheet display. The row(s) stay on your screen when you move to another part of your worksheet. This lets you keep report headings, etc., visible no matter what part of your worksheet you're working on.

To use [LOCK ROW]:

1. Move the cursor to any cell in the bottom row you want to remain on your screen.
2. Press [LOCK ROW] or [ESCAPE] [l] (lower case L). The border of the row the cursor is on, and all rows above it, change to dim display.
3. You can move around your worksheet and edit cells as usual. When you want to unlock the row, press [LOCK ROW] again.

If you try to move the cursor into a locked row, the screen scrolls so the row displays twice. You can change formulas in the "unlocked" copy of the row, but we suggest you unlock a row before editing any of its cells.

## MACROS

The Macro Record/Playback feature lets you record a series of keystrokes, which you can then play back at any time, while you're editing any AlphaCALC worksheet. Macros can greatly reduce time spent doing repetitive tasks, like reformatting multiple layers in a worksheet or several worksheets.

You can record or play back a macro from anywhere in AlphaCALC: your worksheet display, the Main or Editing Menu, or any command screen. And you can include any AlphaCALC feature in a macro: you can enter data in cells, use Main or Editing Menu or slash commands, and use function key features. Anything you can do in AlphaCALC, you can do in a macro.

### RECORDING A MACRO

1. Be sure to have the cursor where you want it to be when you start the procedure you're going to record. This can be anywhere in AlphaCALC.
2. Press [MACROS] or [ESCAPE] [@].

   Your system may be set up with other options from the [MACROS] key. If so, choose macros from the displayed menu.

3. Choose the Record option.
4. Type the name you want to give the macro. This will be used as a file name, so it can be up to six characters long.

5. When you press [EXECUTE] after typing the name, you return to where you were when you called up macros. Now, perform the procedures you want to record. Every keystroke you make is recorded in the macro file.

6. When you're finished recording, press [ESC], followed by [CTRL] / [_] (underline). This ends the recording.

Each macro is stored in a file in the current directory, with the file name you chose and the extension .TIP. Macros are not worksheet specific—you can use any macro in this account in any worksheet in this account.

### PLAYING BACK A MACRO

It's a good idea to **SAVE** the current worksheet before running a macro. If you request the wrong macro name, or start the macro with the cursor in the wrong cell, it could do a great deal of damage to your data. Once you press [EXECUTE] to begin playing back the macro, there is no way to stop it.

1. Be sure to have the cursor where you want it to be when the macro playback begins.

2. Press [MACROS] or [ESCAPE] [@].

Your system may be set up with other options from the [MACROS] key. If so, choose macros from the displayed menu.

3. Select the Playback option.

4. Type the name of the macro you want to use.

5. If you want to perform the actions recorded in the macro more than once, set the iteration field to the number you want.

6. Press [EXECUTE].

You return to where you were when you called up macros, and AlphaCALC performs all the keystrokes recorded in the macro file.

## MODE

The **MODE** command lets you make temporary changes to some of the settings in your AlphaCALC initialization file. The settings you change in **MODE** stay in effect until you leave AlphaCALC.

To use **MODE**:

1. Select it from the Main or Editing Menu, or use **/WM** from your worksheet display.

2. Change the settings you want, then press [EXECUTE].

The settings you can change are:

- Overflow Character: the character displayed when a value is too large to fit in its worksheet cell.
- Absolute Indicator: the character you use to indicate a cell reference in a formula is absolute, not relative.
- Layer Separator: the character used after the layer number in a cell ID.
- Range Separator: the character(s) used between the cell IDs in a range specification.
- Maximum Graph Data Points: the largest number of points allowed in a single data range on a graph.
- Row-Column Border Colors: the foreground and background colors of the row and column borders on your display.

## MOVE

[MOVE] lets you move one or more rows or columns to another location in your worksheet. It affects only entire rows or entire columns, not other ranges. It affects only the current layer. You cannot [MOVE] a row or column from one layer to another.

When you use [MOVE], the row or column is copied to the new location, then deleted from its old location, and other rows or columns are moved to fill the space. Be sure to account for this shifting of parts of your worksheet when choosing the destination.

[MOVE] doesn't recalculate the values of the cells you move. You may want to use [RECOMPUTE] after [MOVE].

To move one or more adjacent rows:

1. To move one row, move to cell A of the row and press [BLOCK MARK].

   To move more than one row, mark cell A of the top and bottom rows you want to move.

2. Move the cursor to cell A of the row you want the moved row(s) to be above.

3. Press [MOVE] or [ESCAPE] [M].

To move one or more adjacent columns:

1. To move one column, move to cell 1 of the column and press [BLOCK MARK]

   To move more than one column, mark cell 1 of the left and right columns you want to move.

2. Move the cursor to cell 1 of the column you want the moved column(s) to be to the left of.

3. Press [MOVE] or [ESCAPE] [M].

## NEXT MATCH

You use the [NEXT MATCH] key during the **SEARCH** command to find the next occurrence of the string you're searching for.

Any time after using **SEARCH**, press [NEXT MATCH] or [CTRL] / [X] to go to the next occurrence of the search string. [NEXT MATCH] continues to work until you leave the worksheet or use **REPLACE** or **GLOBAL**.

When looking for the "next" occurrence, [NEXT MATCH] goes in the last direction you set using [AUTO-ADVANCE]. If you've never used [AUTO-ADVANCE] in this worksheet it goes from left to right, then down to the next row, then in to the next layer.

## NEXT SCREEN

Refer to "Moving Around the Worksheet" in Part 1.

## NEXT WORD

Refer to "Moving Around the Worksheet" in Part 1.

## PAGE BREAKS

The page break feature lets you insert page breaks at specific places in your worksheet, regardless of the page size settings in the Worksheet Header.

The page breaks you insert affect only the printed worksheet, and only when you print by page, not when you print by cell range.

You can insert page breaks between rows and/or between columns. To insert a page break:

1. Move the cursor to the row or column you want to insert a break before. The row or column the cursor is on will be the first of a new page when you print.

2. Type **/PB**. You can use this feature only with a slash command.

3. To insert a page break before this row and/or before this column, set one or both of these fields to Yes.

4. If you want to have your entire worksheet repaginated (as in the **PAGINATE** command), seeing how other page divisions are affected by the new page break, set that field to Yes.

5. Press [EXECUTE].

The page breaks you create show on your worksheet display as a double bar in the row or column border.

To delete a page break:

1. Move the cursor to the row or column with the page break.

2. Type **/PB**.

3. Set the row or column page break (whichever you want to delete) to No. If you don't want a page break at this row or this column, set both fields to No.

4. If you want to repaginate your worksheet, set that field to Yes.

5. Press [EXECUTE].

## PAGINATE

**PAGINATE** divides your worksheet into pages for printing, based on the page format settings in the Worksheet Header and any page breaks you've inserted.

You do not have to use **PAGINATE** before you print—the **PRINT** command paginates before printing your worksheet. However, you can use **PAGINATE** to see where page breaks will occur before using **PRINT**.

Page breaks show on your screen as a single or double bar in the row or column border. The double bar indicates a break you've inserted using the page break feature.

Page breaks between columns affect your worksheet only when you print by page, not when you print by cell range. Inserted page breaks also take affect only when you print by page.

When you print by cell range, pages may be divided at different locations than when you use **PAGINATE**. There is no way to see ahead of time where pages will be broken when printing by cell range. See the **PRINT** command discussion of page format.

To use **PAGINATE**, either choose it from the Editing Menu or type **/PP** from your worksheet display.

Pages are numbered horizontally, then vertically, like this:

| | |
|---|---|
| 1 | 2 |
| 3 | 4 |

All the pages on layer A are numbered in this fashion, followed by layer B, and so on.

## PASTE

**PASTE** copies another worksheet into your worksheet at any position you want. It is often used after **CUT** to move information from one worksheet to another.

To use **PASTE**:

1. Move the cursor to the top left cell of the area you want to copy to. If you are copying a worksheet with multiple layers, the cursor must be in the first layer you want to copy to.

2. Select **PASTE** from the Editing Menu, or type **/FP** from your worksheet.

3. Type the name of the worksheet you want to copy. If it has a password, you'll have to enter it.

3. Press EXECUTE.

The worksheet you chose is copied into yours, starting at the cursor position.

Any existing information in the cells you're copying to is erased, even if the cells were protected.

## PREVIOUS SCREEN

Refer to "Moving Around the Worksheet" in Part 1.

---

## PREVIOUS WORD

Refer to "Moving Around the Worksheet" in Part 1.

---

## PRINT

The **PRINT** command lets you print all or part of your worksheet to a printer or a disk file.

You see two screen displays while using **PRINT**. The first lets you choose how you want your worksheet printed: by page or cell range, and so forth. The second contains options relating to the printing process: which printer, how many copies, and so on.

The format of your printed worksheet is largely determined by the page format settings in the Worksheet Header. See the discussion of page format below.

To use **PRINT**:

1. Select it from the Main Menu or Editing Menu, or, from your worksheet display, type **/PW**.

2. If you used **PRINT** from the Main Menu, choose the worksheet you want to print. If it has a password, you have to enter it, also.

3. **PRINT** paginates your worksheet, then you see the worksheet printing options screen. You can set the following fields:

   - Output Format: do you want to print cell data (contents as shown in the worksheet cells), cell formulas, or both? If you print cell formulas, or both data and formulas, you must print by cell range, not by page.

   - Print by ...: do you want to print a range of cells, or a number of pages?

   - Cell Range: this field appears only if you’re printing by cell range. Type the starting and ending cells you want.

   - Print Page(s): only when you’re printing by page, enter the starting and ending page numbers. Pages are numbered horizontally, then vertically, as described in **paginate**.

4. When you’ve chosen the options you want, press [EXECUTE].

5. You now see the printer selection screen. You can choose the printer you want, and set these options:

   - Send Output to the Printer?: Do you want to print the worksheet now, or save it as a disk file? If you save it, the file has the same name as your work sheet, and the first available extension in the series .000, .001, and so on.

   - Erase Output After Printing?: Do you want the print file erased from the disk after it prints? If you don't erase the file, it is named as described above.

   - Forms Type: What form name should the worksheet print on? This lets you change the default form name for this worksheet in the Worksheet Header. If the printer you choose has a different form mounted, your worksheet won't print until the proper form is put on.

   - Number of Copies: You can print up to 999 copies.

6. When you've set the printer options, press [EXECUTE].

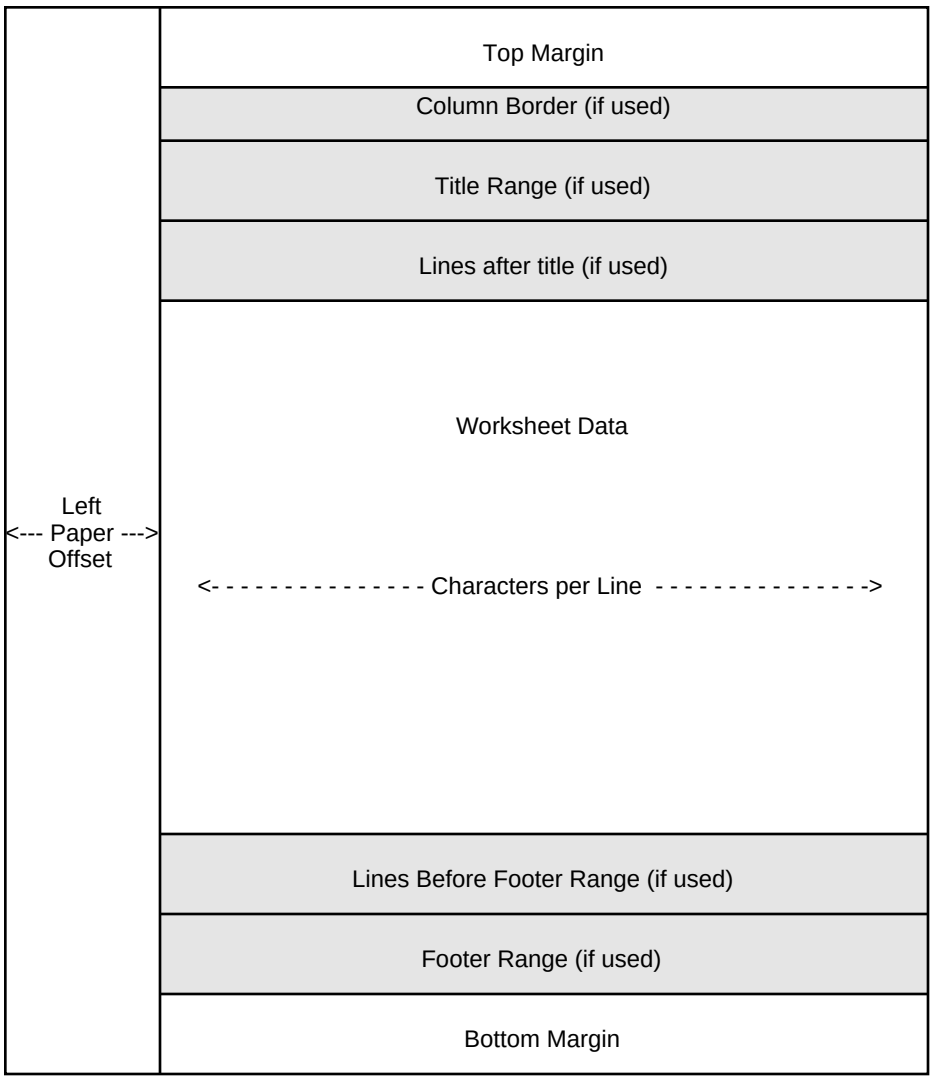
## PRINTED PAGE FORMAT

The page format of your printed worksheet is determined by three things:

- The page format settings in the Worksheet Header.
- Whether you're printing by cell range or by page.
- If you're printing by page, any page breaks you've inserted using /PB.

### Printing by Page

When you print by page, the format of a printed page looks like this:




The total number of lines from the top of the top margin to the bottom of the bottom margin equals the Page Length setting in the header. This is normally 66 for 11" paper.

To determine how many rows of your worksheet print, add the lines used for all the other sections shown in the diagram, and subtract the total from the Page Length setting. For example, if your Worksheet Header settings are as follows:

| | |
|---|---|
| Page Length | 66 |
| Top Margin | 3 |
| Bottom Margin | 3 |
| Title Range | A1:G2 (2 lines) |
| Lines After Title | 1 |
| Footer Range | None |
| Column Border? | Yes |

The number of lines of your worksheet per page is:

66 - (3 + 3 + 2 + 1 + 1) = 66 - 10 = 56

If you include the title range in the page range you print, it will print twice, once as a title and once in the Worksheet Data area. To avoid this, put the title range on a page by itself (if it's at the top of your worksheet, you can use the page break feature to put a page break after it), and don't include that page in your print request. This is also true for the footer range.

As shown in the diagram, the page width is the total of the Left Paper Offset and Characters per Line settings. Both of these settings are in characters. If you're printing ten characters per inch on 8.5" wide paper, these fields should add to about 80 (leaving a half-inch right margin).

If you are printing the row border, it takes up the first six characters of each line in the Worksheet Data area.

AlphaCALC compares the Characters per Line setting with the column widths in your worksheet—it will not split a page in the middle of a column.

Of course, "hard" page breaks you've inserted using /PB will override the Worksheet Header settings, and may result in smaller pages than described here.

If you want to fit more columns of your worksheet on one page, and your printer is capable of it, you can print in landscape mode. However, you must remember to change the header format settings if you do this, since the paper length is now 8.5" and the width is 11" instead of the other way around.

### Printing by Cell Range

Page format is much simpler when printing by cell range:

- Inserted page breaks are ignored.
- Column page breaks are ignored. Each row of your worksheet prints on one line.
- The title range and footer range are ignored. You cannot print titles and footers when you're printing by cell range.

So, the number of rows on each page equals: Page Length - (Top Margin + Bottom Margin + 1 if printing column border).

The Left Paper Offset setting still determines the left margin, but the entire row is printed across the page, regardless of its length.

## PROTECT

The **PROTECT** command lets you protect, or remove protection from, a range of cells. To protect a single cell, you can use [PROTECT CELL]. You cannot edit or erase protected cells, and many other commands, such as **FORMAT**, **REPLACE**, and **GLOBAL**, won't affect them. Also, [AUTO ADVANCE], [TAB], [NEXT WORD], [BACK TAB], and [PREV WORD] skip protected cells.

These features can delete or copy over protected cells: [BLOCK COPY], [DELETE COLUMN], [DELETE ROW], **DUPLICATE**, **PASTE**, and **IMPORT**.

To use **PROTECT**:

1. Select it from the Editing Menu, or type **/RP** from your worksheet display.
2. Enter the cell range you want. If there is a marked range in your worksheet, it displays as the default.
3. Choose whether to protect the cells or remove protection from them.
4. Press [EXECUTE].

When the current cell is protected, its cell ID at the bottom left of your screen is shown in brackets. For example, [A1].

## PROTECT CELL

The [PROTECT CELL] key lets you protect a single cell, or remove protection from it, without leaving your worksheet. To protect multiple cells, use the **PROTECT** command. You cannot edit or erase protected cells, and many other commands, such as **FORMAT**, **REPLACE**, and **GLOBAL**, won't affect them. Also, [AUTO ADVANCE], [TAB], [NEXT WORD], [BACK TAB], and [PREV WORD] skip protected cells.

These features can delete or copy over protected cells: [BLOCK COPY], [DELETE COLUMN], [DELETE ROW], **DUPLICATE**, **IMPORT**, and **PASTE**.

To use [PROTECT CELL]:

1. Move the cursor to the cell you want to protect or remove protection from.
2. Press [PROTECT CELL] or [ESCAPE] [p].

If the cell was not protected, now it is; if it had been protected, now it isn't.

When the current cell is protected, its cell ID at the bottom left of your screen is shown in brackets. For example, [A1].

## QUEUE

The **QUEUE** command lets you look at and modify the printer queue—the list of files waiting to print on a particular printer.

Besides checking the progress of printing files, you can do the following with **QUEUE**:

- Move a file to the top of the queue, so it prints immediately after the current file finishes.
- Remove a file from the queue.
- Change the name of the form mounted on the printer. You need to do this if a file in the queue prints on a different type of form than the one currently mounted.

To use **QUEUE**:

1. Select it from the Main Menu or Editing Menu, or type **/PQ** from your worksheet display.
2. Select the printer whose queue you want to see.

   The queue display shows the files in the order they will print, including: a file number assigned by AMOS, the file name, number of copies requested, form name for the file, and the number of blocks to print for the file.
3. To use one of **QUEUE**'s options, select the option, then type the number of the file you want to affect (for Remove or Select), or the name of the new form you're mounting. Press [EXECUTE].
4. When you've made all the changes you want, and are through viewing the queue display, press [MENU].

## QUIT - Editing Menu

The Editing Menu **QUIT** command takes you from your worksheet display back to the Main Menu. It does ***not*** save your worksheet to the disk. When you use **QUIT**, your worksheet returns to how it was when you last used either **SAVE** or **FINISH**.

If you use **QUIT** to leave a worksheet you've just created, and have never used **SAVE** or **FINISH** with, the entire worksheet is erased.

To use **QUIT**:

1. Select it from the Editing Menu or type **/Q** from your worksheet display.
2. If you've made changes or additions to your worksheet since calling it up or saving it, AlphaCALC asks if you are sure you want to quit and lose your changes. If you are, type **YES** [RETURN].

## QUIT - Main Menu

The **QUIT** command from the AlphaCALC Main Menu takes you out of AlphaCALC. You return to AMOS command level, or to the menu from which you called up AlphaCALC.

To leave AlphaCALC, from the Main Menu, select **QUIT**.

---

## RECOMPUTE

The [RECOMPUTE] key recalculates the values of all the formulas in your worksheet. You need to use [RECOMPUTE] to update your worksheet after changing any values which are referred to by other cells.

To use [RECOMPUTE], press it or [ESCAPE] [R] any time you are at your worksheet display.

If you have any circular reference formulas in your worksheet, you may want to use the iteration option with [RECOMPUTE], described below.

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## RECOMPUTE - with iteration

To perform some calculations, you may want to use a circular reference cell formula—a formula which directly or indirectly refers to itself. Such a formula may change its value each time you recompute it.

In this case, you can use the [RECOMPUTE] iteration option, which lets you recalculate your worksheet formulas a specified number of times, or until a value you choose changes by less than a given amount with each iteration.

Only cell formulas with circular references, and cells which refer to them, are affected by this option. Other parts of your worksheet will not change after the first iteration.

Because this option is designed to be used with circular references, you will not see any "Circular reference" error messages when you use it.

To use [RECOMPUTE] with iteration:

1. Press [ITERATION] or [ESCAPE] [r].
2. On the Iteration Options screen, set the following values:
   - Recalculate by: should the formulas be computed row by row, column by column, or layer by layer?
   - From: should the formulas at the top or bottom of the worksheet be computed first?

- Iteration Count: how many times should each formula be evaluated? The maximum is 255.
- Delta: if the value of the cell you specify changes less than this much on any iteration, the recomputation will stop. This field cannot be zero.
- Cell: which cell should be checked for the Delta change amount?

3. When you've set all the fields you want, press [EXECUTE].

AlphaCALC now recalculates each cell in the worksheet the number of times you chose, or until the cell you specified changes by less than the Delta amount, whichever comes first.

## RENAME

The **RENAME** command lets you change the name of any worksheet in this account.

To use **RENAME**:

1. Select it from the Main Menu.
2. Type the name of the worksheet you want to rename, or use [↑] and [↓]. to display the worksheet directory and select the worksheet you want from the list.
3. If the worksheet has a password, type it.
4. Type the name you want to give the worksheet.
5. Press [EXECUTE].

## REPLACE

The **REPLACE** command searches your worksheet—or a cell range—for a group of characters, and gives you the option of replacing each occurrence with the characters you specify.

If you just want to look for a character string, without necessarily changing it, use **SEARCH.**

**REPLACE** can search all of your cell entries or just certain types. Your choices are:

- Constants: only text entries and numbers. This includes only entries where nothing is displayed in the value field.

- Expressions: math, logical, or string expressions—any cell where AlphaCALC evaluates your entry and displays the calculated value in the cell. **REPLACE** searches what you entered in the cell, not its displayed value.

- Formulas: all cell formulas, both constants and expressions.

For example: you enter 4 in one cell, and 4+3 (which displays as 7) in another. If you use **REPLACE** to replace 4 with 6, searching constants only, only the first cell is changed. If you search expressions, only the second cell is modified—it now equals 6+3, or 9.

Constants and expressions are discussed in "Types of Cell Formulas" in Part 1.

To use **REPLACE**:

1. Select it from the Editing Menu, or type **/WR** from your worksheet display.

2. The default range is either your entire worksheet or the currently marked block of cells. Type the range you want if this is not correct.

**REPLACE** will not change protected cells.

3. Type the string you want to look for, and the one you want to replace it with.

4. Choose constants, expressions, or formulas, as discussed above.

5. You can choose to make this a global replacement. If you change the `Global Replace?` field to **Y**, every occurrence of the string is replaced, as in the **GLOBAL** command.

5. When you've made all your entries and choices, press [EXECUTE].

Unless you chose global replacement, you now see the first occurrence of the search string, and the prompt:

```
Replace? (Yes, No, Quit):
```

Type the first letter of your choice: **Y** to replace the string, **N** not to, or **Q** to end the **REPLACE** command. Unless you chose quit, **REPLACE** then goes to the next occurrence of the string, and so on.

When looking for the string, **REPLACE** goes in the last direction you selected using [AUTO-ADVANCE]

## SAVE

The **SAVE** command stores your worksheet on the disk. Until you use **SAVE** (or **FINISH**), your changes and additions are stored only in the computer's temporary memory, and would be lost in the unlikely event of a power failure, someone rebooting the computer, or the computer going down for some other reason.

For this reason, we suggest you use **SAVE** at regular intervals while working in AlphaCALC. It's also a good idea to **SAVE** your worksheet before making changes you may not want to keep, such as a major reformatting. That way, if you decide you like the old format better, you can **QUIT** to return to it.

To use **SAVE**, select it from the Editing Menu, or type **/FS** from your worksheet display.

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## SCREEN LEFT

See "Moving Around Your Worksheet" in Part 1.

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## SCREEN RIGHT

See "Moving Around Your Worksheet" in Part 1.

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## SEARCH

**SEARCH** lets you look through your worksheet, or part of it, for the character string you specify.

If you want to replace a character string with another string throughout your worksheet, you can use **REPLACE** or **GLOBAL**.

You can search all cell formulas and values, or only certain types. You have these choices:

- Constants: only text entries and numbers. This includes only entries where nothing is displayed in the value field.
- Expressions: math, logical, or string expressions—any cell where AlphaCALC evaluates your entry and displays the calculated value in the cell.
- Formulas: all cell formulas, both constants and expressions.

- Values: what is displayed in the cell. For constants, this is the same as the formula; for expressions, it is the value you get when you evaluate the expression.

- All: all types of cell formulas, and cell values.

For example, if you have 11 entered in one cell, and 6+5 in another, and search for 11:

- You'll find only the first cell if you specified constants or formulas.
- You'll find both cells if you search values or all.
- You won't find either cell if you search expressions.

Constants, expressions, formulas, and values are discussed in a little more detail in Part 1.

To use **SEARCH**:

1. Select it from the Editing Menu, or type **/WS** from your worksheet display.

2. The default cell range is either your entire worksheet or the currently marked cell range, if there is one. If you want to search a different range, type it.

3. Type the string you want to look for.

4. The remaining fields let you set these options:

   Look in: what types of entries do you want to search, as discussed above?

   Partial Search: do you want it considered a match if your string is part of a cell entry, or only if your string exactly matches the entire cell formula? For example, one cell formula is A5, another is A5/100, and you search for A5. A partial search will find both occurrences, while if partial search is off, only the first cell is a match.

   Searchfold on: do you want **SEARCH** to consider the case of text when looking for your string, or should it match both upper and lower case, regardless?

5. Once you've set the options you want, press [EXECUTE].

6. You return to your worksheet display, with the cursor at the first occurrence of the string. To go to the next occurrence, press [NEXT MATCH].

[NEXT MATCH] always takes you to the next occurrence of the string you searched for. You can edit cells or use any other AlphaCALC feature, and [NEXT MATCH] will still work. Only returning to the Main Menu or using **REPLACE** or **GLOBAL** will cancel the search.

[NEXT MATCH] goes in the last direction you set using [AUTO ADVANCE]. If you've never used [AUTO ADVANCE] in this worksheet, it goes to the right, then down, then to the next layer.

## SORT

**SORT** lets you put your worksheet, or a cell range you select, in order according to the contents of one, two, or three columns of cells.

**SORT** orders the rows in the range you choose based on the values of the cells in the column(s) you select. For an ascending sort, **SORT** follows these rules:

- First, the cells are put in order according to type of value: numbers, text, logical, blank.
- Within each type, they are put in alphanumeric order: 0 - 9, A - Z, a - z.

A descending sort reverses this order.

You can choose up to three columns as sort keys, and can select ascending or descending order for each. The keys are used in this order:

- The primary key values for each row are compared.
- If any rows have identical primary keys, they are sorted according to their secondary keys.
- Tertiary keys are used only if both the primary and secondary key values are the same.

When rows are moved during **SORT**, relative cell references are adjusted as they are when you copy a cell. If cells inside the sort range refer to cells outside the sort range, be sure to take this adjustment into account.

To use **SORT**:

1. Select it from the Editing Menu, or type **/RS** from your worksheet display.
2. Enter the cell range you want to sort. Be sure to include all the cells you want to move as a result of the sort, not just the columns you're using as sort keys.

   The sort range must be all on one layer; you can't enter a three-dimensional range.

   If there is a marked block in your worksheet, it is the default **SORT** range.
3. You must specify a primary key. The secondary and tertiary keys are optional. For each key you use, you can set ascending or descending order.

   For any key column which contains text entries, you can choose to sort based on only part of the text. To do so, enter the number of characters you want to sort on. **SORT** always begins with the first character in each cell.

You can also choose to fold characters to upper case for the sort. If you don't, in an ascending sort, all upper case characters are sorted before lower case, as described above.

4. When you've selected all the options you want, press [EXECUTE].

## SPEED MODE

The speed mode feature is a way for you to speed up entering data in your worksheet. When speed mode is turned on, the [RETURN] and [ENTER] keys work just like [EXECUTE] after you enter a cell formula.

This is very useful if you're entering a lot of data using the ten-key pad on your keyboard. By using [ENTER] instead of [EXECUTE], you can enter all the numbers without moving your hand from the ten-key pad, especially if you also use [AUTO ADVANCE] to move to the next cell after each entry.

When you're in speed mode, you can still go to the cell format fields by pressing [↓] or [TAB] after entering the cell formula. From any of the format fields, you must press [EXECUTE] to finish editing the cell—speed mode affects only the cell formula field.

To turn on speed mode, press [ESC] [CTRL] / [S].

To turn speed mode off when it is on, press [ESC] [CTRL] / [S] again.

When speed mode is on, an S displays at the right-center of the bottom status line.

## WIDTH

The **WIDTH** command changes your worksheet display between 80 column and 132 column mode. It affects only your worksheet display, not any menus or other screens.

**WIDTH** works only if your terminal supports both 80 and 132 column display.

To use **WIDTH**, select it from the Editing Menu, or type **/WW** at your worksheet display.

## WRITE

**WRITE** lets you take all or part of your worksheet and create an AlphaWRITE document from it. You can then use AlphaWRITE to format your worksheet for printing, include it in reports written in AlphaWRITE, and so on.

**WRITE** copies only cell values to AlphaWRITE; cell formulas are not transferred.

**WRITE** creates AlphaWRITE 1.2A format documents. If you use a later version of AlphaWRITE, the documents will be converted to proper format when you call them up.

Even if your worksheet is password protected, the AlphaWRITE document you create does ***not*** have a password. Also, AlphaCALC does not put any page breaks in the AlphaWRITE document.

To use **WRITE**:

1. Select it from the Editing Menu or type **/FW** from your worksheet display.

2. Type the name you want the created document to have. You can view a document directory by pressing [↑] or [↓].

3. You can specify either a cell range or a page range to include in the document. Choose which method you want, then specify either the range or the starting and ending page numbers.

   If you have marked a block of cells and choose to transfer a cell range, the marked block is the default range.

4. Press [EXECUTE].

# PART THREE
# FUNCTIONS

A function is a shorthand way to perform a mathematical or other operation. You enter the function, and AlphaCALC understands it as "perform this operation." Function calculations range from simple, such as adding a group of numbers, to complex, such as looking up a value in a table and returning a corresponding value from another cell.

When you enter a function, you include the function name and one or more arguments:

- The function name tells AlphaCALC what you want it to do.
- The argument(s), which are enclosed in parentheses after the function name, tell AlphaCALC what value or values you want the function to use. A few functions don't have any arguments.

For example, in the function SUM(D5:F12), the function name SUM tells AlphaCALC to add a group of numbers, and the cell range D5:F12 tells it which values to add.

After a few general notes about function arguments, this part of the *Reference Book* describes most AlphaCALC functions. Some of the more technical functions, such as changing the internal data type of a stored value, are covered in the *AlphaCALC Programmer's Information Manual*.

## FUNCTION ARGUMENTS

The descriptions of the arguments for each function should be fairly clear. Keep these general rules in mind when entering a function argument:

- A "numeric expression" is anything with a numeric value. It could be a number, a cell ID, another function, a combination of these, and so on.
- A "string" can be either a string constant (where you actually type in the characters) or an expression having a string value (such as the cell ID of a cell containing a string). If you use a string constant, it must be enclosed in quotation marks, like this: "STRING".
- A "cell range or list" is a combination of one or more single cell IDs and/or cell ranges, separated by commas. For example, A3:F4,G2,L6.

- When the argument for a date function is listed as *date*, such as *date 1*, it is a date string. A date string must be in the correct format for your language. For American English, the format is MM/DD/YY. The date can either be a literal string, such as "04/23/91", or an expression, such as a cell reference or another date function, which has a value of a date string. If it is a literal string, you must enclose it in quotes, as shown here.

  Other date functions require you to enter a "serial date," which is the number of days from January 1, 1900 to the date you want. If you don't know the serial date, you may be able to use another date function to find it.

## ABS

ABS(*numeric expression*)

The ABS function displays the absolute value of the value of its argument. The absolute value is the unsigned value. For example, ABS(-14.5) = 14.5.

## ACOS and ACOSD

ACOS(*numeric expression*)
ACOSD(*numeric expression*)

The arccosine functions, ACOS and ACOSD, find the arccosine of the value of their arguments. The arccosine is the angle which has a cosine of that value.

ACOS displays the arccosine in radians between 0 and PI.

ACOSD expresses the arccosine in degrees from 0 to 180.

To find a cosine, use the cosine functions, COS and COSD.

## ACUM

ACUM [RETURN]; *value*; [EXECUTE]; repeat; when done, [RETURN], then [EXECUTE]

ACUM is different than any other AlphaCALC function. It turns a cell into a simple adding machine. After entering the ACUM function, you type a series of numbers, pressing [EXECUTE] after each. The running total of the numbers appears in the cell. When you've added all the numbers you want, press [RETURN] to leave the Add value field. You can then set the format fields as with any cell formula, and press [EXECUTE] when you're done.

Pressing MENU or CANCEL in the `Add value` field clears the accumulator and returns you to the `Formula` field.

To clear an accumulator, type **`C`** in the `Add value` field.

## AND

`AND(`*`logical expression,logical expression`*`)`

AND checks to see if two logical expressions are true. If both expressions are true, AND is true. If one or both expressions are false, AND is false.

You can also use AND as an operator, as described in Part 1.

## ASC

`ASC(`*`string`*`)`

The ASC function returns, as a number, the ASCII value of the first character of *`string`*. All other characters are ignored. For example, `ASC("ABC")` = 65.

## ASIN and ASIND

`ASIN(`*`numeric expression`*`)`
`ASIND(`*`numeric expression`*`)`

The arcsine functions, ASIN and ASIND, find the arcsine of the value of their arguments. The arcsine is the angle which has a sine of that value.

ASIN displays the arcsine in radians between -PI/2 and PI/2; ASIND expresses the arcsine in degrees from -90 to 90.

To find the sine of an angle, use SIN or SIND.

## ATAN and ATAND

`ATAN(`*`numeric expression`*`)`
`ATAND(`*`numeric expression`*`)`

The arctangent functions, ATAN and ATAND, find the arctangent of the value of their arguments. The arctangent is the angle which has a tangent of that value.

ATAN displays the arctangent in radians between -PI/2 and PI/2.

ATAND expresses the arctangent in degrees from -90 to 90.

To find an angle's tangent, use TAN or TAND.

---

## AVERAGE

`AVERAGE(`*`cell range or list`*`)`

The AVERAGE function computes the average of the values of the cells in its argument—it totals the values in the cells and divides by the number of cells.

Cells with a value of zero are included in the calculation; blank cells and cells containing text are not.

---

## CMONTH

`CMONTH(`*`numeric expression`*`)`

CMONTH returns the name of the month specified by *`numeric expression`*. The value of *`numeric expression`* must be an integer from 1 to 12. For example, CMONTH(3) = March.

---

## COLS

`COLS(`*`cell range`*`)`

COLS counts the number of columns in *`cell range`*. For example, COLS(G5:K16) equals five.

## COS and COSD

`COS(`*`radians`*`)`
`COSD(`*`degrees`*`)`

The two cosine functions, COS and COSD, compute the cosine of the angle specified in their arguments. If the size of the angle is expressed in radians, use COS; if it is expressed in degrees, use COSD. Both *`radians`* and *`degrees`* can be any numeric expression.

You can find the arccosine of a cosine value by using ACOS or ACOSD.

---

## COUNT

`COUNT(`*`cell range or list`*`)`

COUNT counts the number of cells in its argument which have numeric or logical values. Cells with a value of zero are counted; blank cells and cells containing text are not.

---

## COUNTA

`COUNTA(`*`cell range or list`*`)`

COUNTA counts the number of non-blank cells in its argument. Cells with any numeric, logical, or string value are counted; blank cells are not. COUNTA differs from COUNT in that it counts cells containing logical or string values.

---

## COUNTV

`COUNTV(`*`value,cell range or list`*`)`

COUNTV counts the occurrences of `value` in the specified cell range. For example, `COUNTV(5,D12:G26)` returns the number of cells in the range with a value of 5.

COUNTV counts matches with cell values only, not cell formulas. In the above example, a cell formula of F6/5 is not counted, while a formula of 10/2 is.

`Value` can be any type of numeric expression; it cannot be a string.

## CTERM

`CTERM(`*`i,endval,startval`*`)`

- *`i`* is the interest rate, expressed as a decimal.
- *`endval`* is the ending value you want.
- *`startval`* is the starting value.

The CTERM financial function figures how many periods it takes for a current value (*`startval`*) to increase to a given value (*`endval`*) at a specified interest rate (*`i`*).

Be sure to express the interest rate for the correct period. CTERM assumes interest is compounded once per period, so if interest is compounded monthly, *`i`* must be the monthly interest rate. In this case, the value of CTERM will be the number of months it takes for *`startval`* to increase to *`endval`*.

All three parts of the argument can be any type of numeric expression.

CTERM is similar to RATE, which figures the interest rate necessary for an investment to increase in a specified amount of time.

---

## DATE

`DATE(`*`YY,MM,DD`*`)`

DATE returns the serial date—the number of days since January 1, 1900—of the date you enter. You must enter the numbers for *`year, month`* and *`day`* in the order shown. They do not need to be in quotes. For example, `DATE(57,3,24)` for March 24, 1957.

---

## DATEVALUE

`DATEVALUE(`*`date`*`)`

DATEVALUE computes the serial date for the date in its argument. The serial date is the number of days elapsed since January 1, 1900.

*`date`* is a date string. It must be in the correct format for your language (MM/DD/YY for American English). If it is a literal string, it must be enclosed in quotes.

## DAY

`DAY(`*`serial date`*`)`

The DAY function figures out the day of the month corresponding to the serial date in its argument. The serial date is the number of days since January 1, 1900. For example, if *`serial date`* is 385 (equal to January 21, 1901), then DAY equals 21.

MONTH and YEAR let you extract the month number or year from a serial date.

---

## DAYS

`DAYS(`*`date 1,date 2`*`)`

DAYS counts the number of days between *`date 1`* and *`date 2`*. If *`date 1`* is larger than *`date 2`*, *`date 2`* is assumed to be in the next century.

Both dates are date strings. They must be in the correct format for your language (MM/DD/YY for American English). If they are literal strings, they must be enclosed in quotes.

You can count the weeks or months between two days by using WEEKS or MONTHS.

---

## DDB

`DDB(`*`cost,salvage,life,year)`*`)`

The DDB function calculates depreciation using the double declining balance method. You must provide the cost of the item, its salvage value, the useful life of the item (the number of years over which it depreciates in value from *`cost`* to *`salvage`*), and the year you want the depreciation amount for.

Each part of the argument can be any numeric expression.

For example, if an item costs $25,000, and depreciates to a salvage value of $500 over 15 years, you can figure its depreciation in the eighth year with the formula:
DDB(25000,500,15,8) = $1224.17

The SLN and SYD functions calculate depreciation using other methods.

## DING

`DING(`*`numeric expression`*`)`

The DING function lets you ring your terminal bell a specific number of times. DING figures the value of the expression in its argument, and rings the terminal bell that many times.

The value of DING's argument must be an integer from 0 to 255.

---

## EXP

`EXP(`*`numeric expression`*`)`

EXP, the exponential function, calculates ***e*** —the base of the natural logarithms, approximately 2.71828—raised to the power of the value of its argument.

LN is the reverse of EXP. It finds the natural logarithm of a numeric value.

---

## EXTCELL

`EXTCELL(`*`worksheet name,cell ID`*`)`
`EXTCELL(`*`worksheet name,function(cell range)`*`)`

The external cell function, EXTCELL, lets you take a value from another worksheet and use it in this worksheet. There are two ways to use EXTCELL:

- You can copy the value of a single cell. For example: EXTCELL(TAXES,H24).
- You can perform an arithmetic function on a range of cells in the other worksheet, and transfer the value of the function to this cell. You can do this only with these functions: AVERAGE, COUNT, COUNTA, MAX, MIN, SUM, STDEV, STDEV1, VAR, and VAR1. For example:

```
EXTCELL(FEBSAL,SUM(L45:L57))
```

## FACT

`FACT(`*`numeric expression`*`)`

FACT calculates the factorial of the value of its argument. The factorial is the product of all the integers from 1 to the value. For example, FACT(5) equals 1*2*3*4*5, or 120.

The value of *`numeric expression`* must be a positive integer.

---

## FALSE

`FALSE`

The FALSE function always has the logical value of FALSE. It has no argument.

FALSE can be very useful in checking the construction of formulas using other logical functions, especially complex IF functions. By substituting FALSE for the expression in an IF argument, you can make sure the result for the second (false) branch of the IF function is correct.

---

## FISCALMONTH

`FISCALMONTH(`*`numeric expression`*`)`

FISCALMONTH figures the current month of the fiscal year. The value of its argument is the starting month for the fiscal year you are concerned with. This value must be an integer: 0 or 1 equals January, 2 = February, and so on. Any number 12 or above equals December.

FISCALMONTH then uses the system date to figure the month of the fiscal year, based on the starting date in the argument.

---

## FIX

`FIX(`*`numeric expression`*`)`

FIX truncates the value of its argument to an integer. It drops all digits to the right of the decimal point. For example, FIX(-18.64) = -18.

For all positive numbers, FIX and INT give the same result. For negative numbers (except integers), FIX and INT have different values.

## FV

`FV(`*`p,i,n`*`)`

- *p* is the payment per period.
- *i* is the interest rate, in decimal form.
- *n* is the number of periods.

FV, the future value function, calculates the future value of an annuity.

All the values in the argument for FV must be positive. The interest rate must be between 0 and 1, and the number of periods must be an integer.

Be sure all the parts of the argument match. If the annuity has monthly payments, enter the monthly (not annual) interest rate, and the length of the annuity in months.

The NPV and PV functions may also be helpful in examining annuity values.

---

## HLOOKUP

`HLOOKUP(`*`expression,cell range,offset`*`)`

- *expression* is the string or numeric value you want to look for. If it is a string constant, it must be in quotation marks.
- *cell range* is the range containing the table you want to search.
- *offset* determines the row HLOOKUP returns the value from.

HLOOKUP performs a lookup function in a horizontal table. It searches the top row of *cell range* for the value or string you're looking for. If there is no exact match for a numeric value, it finds the next lower value. Then, it moves down the number of rows specified in *offset*, and returns the value of the corresponding cell in that row.

For example, assume this simple table starts in cell A1:

```
2          3          5
WRENCH     PLIERS     SCREWS
$4.95      $1.99      $0.05
```

Here are some HLOOKUP functions and their results:

```
HLOOKUP(3,A1:C3,2)            $1.99
HLOOKUP("SCREWS",A2:C3,1)     $0.05
HLOOKUP(4,A1:C3,1)            PLIERS
```

Notice, in the last example, since there was no exact match for 4, HLOOKUP used the next lower value.

The values in the row you search must be in ascending order, and there should not be any duplicates. If there are duplicates, only the first occurrence will be found.

VLOOKUP performs vertical table lookups; LOOKUP can be used with all kinds of tables.

---

## IF

`IF(`*`logical expression,expression 1,expression 2`*`)`

IF is one of AlphaCALC's most powerful functions. It has the value of *`expression 1`* if *`logical expression`* is true, or the value of *`expression 2`* if *`logical expression`* is false. *`Expression 1`* and *`expression 2`* can be numbers or string constants, or any kind of numeric, logical or string expression. Combined with AlphaCALC's logical operators and other logical functions, this lets you check for positive or negative numbers, exact values or values in a given range, and so on.

You can include other IF functions in the expressions of an IF argument, letting you choose among several final values depending on the values of many different cells. For example:

```
IF(A6>0,IF(B6<4,C6,D6),IF(A8>0,B8,"None"))
```

You can also leave out either *`expression 1`* or *`expression 2`*, and the cell will be blank if that expression would have been displayed. If you choose to leave out *`expression 1`*, you must include two commas before *`expression 2`*. For example:

```
IF(A2=9,,"Not equal")
```

This cell is blank if A2=9, or displays `Not equal` if it isn't equal.

---

## INDEX

`INDEX(`*`cell range,numeric expression 1,numeric expression 2`*`)`

INDEX looks up a specific cell within a table and displays the value of that cell. The three parts of INDEX's argument describe the cell:

- The *`cell range`* is the table you want to look the cell up in. The top left corner of this range sets the 0,0 coordinate for the table.
- The value of *`numeric expression 1`* is the column within the table. The left most column is column 0, the next is column 1, and so on.
- The value of *`numeric expression 2`* is the row number within the table. The top row is 0, the next is 1, and so on.

For example:

```
INDEX(G15:L22,3,5)
```

returns the value of cell J20. Obviously, the numeric expressions would normally be more complex, and the cell value displayed would depend on one or more variable(s).

---

## INT

`INT(`*`numeric expression`*`)`

The INT function displays the largest integer equal to or less than the value of its argument. For example, INT(6)=6 and INT(4.2-16.5)=-13.

Notice that with negative numbers, the next smallest integer is the next "highest" negative number. In our example, -13 is smaller than -12.3.

INT gives the same result as FIX with positive numbers, but their values are different for negative non-integers.

---

## LCS$

`LCS$(`*`string expression`*`)`

The LCS$ string function takes the string in its argument and changes it to all lower case. If `string expression` is a string constant, it must be in quotes.

You can also change the case of strings using PROPER$ and UCS$.

---

## LN

`LN(`*`numeric expression`*`)`

LN calculates the natural logarithm of the value of its argument. The natural log is the log to the base ***e***, a mathematical constant equal to about 2.7818.

The EXP function is the opposite of LN; it calculates ***e*** raised to the power of its argument.

## LOG10

`LOG10(`*`numeric expression`*`)`

LOG 10 displays the common logarithm—the log to the base 10—of the value of its argument.

You can calculate natural logarithms using LN.

---

## LOOKUP

`LOOKUP(`*`numeric expression,cell range`*`)`

The LOOKUP function lets you search any row or column in your worksheet for a specific value, and return the value of the corresponding cell in another row or column.

The cell range you enter for LOOKUP determines three things: whether LOOKUP searches a row or column for the value, what row or column it looks in, and what row or column it returns the corresponding value from.

If the range has at least as many rows as columns, LOOKUP searches the first column for the value, and returns the value from the cell in the same row of the last column of the range.

If the range has more columns than rows, LOOKUP searches the first row, and returns the value from the corresponding cell in the last row.

If the row or column LOOKUP searches does not have an exact match for the value of *`numeric expression`*, it considers the next lowest value a match, and displays the value of the cell corresponding to it.

LOOKUP will not search for or display text strings. You must use either HLOOKUP or VLOOKUP to do so.

---

## MAX

`MAX(`*`cell range or list`*`)`

MAX finds the largest value in any of the cells in the range or list in its argument. It ignores blank cells and text strings, but counts values of zero.

MIN finds the smallest value in a group of cells.

## MIN

`MIN(`*`cell range or list`*`)`

MIN finds the lowest value in any of the cells in its argument. It ignores blank cells and text strings, but counts values of zero. Remember, negative values are lower than a value of zero.

MAX finds the largest value in a group of cells.

---

## MOD

`MOD(`*`numeric expression 1,numeric expression 2`*`)`

MOD finds the modulus (remainder) when the value of *`numeric expression 1`* is divided by the value of *`numeric expression 2`*.

The value of *`numeric expression 2`* should be an integer.

---

## MONTH

`MONTH(`*`serial date`*`)`

MONTH extracts just the number of the month (1-12) from the serial date in its argument. A serial date is the number of days since January 1, 1900.

You can find just the day or year of a serial date using DAY or YEAR.

---

## MONTHS

`MONTHS(`*`date 1,date 2`*`)`

MONTHS calculates the approximate number of months between the two dates in its argument. The number of months is considered to be the number of days divided by 30.

Both dates are date strings. They must be in the correct format for your language (MM/DD/YY for American English). If they are literal strings, they must be enclosed in quotes.

If *`date 1`* is smaller than *`date 2`*, it is considered to be in the next century.

## NOT

`NOT(`*`logical expression`*`)`

NOT equals the opposite of the logical value of its argument. If the expression in the argument is true, NOT is false, and vice versa.

---

## NPV

`NPV(`*`i,cell range or list`*`)`

- *`i`* is the interest rate per period, in decimal form.
- *`cell range or list`* contains the amounts to be received, in order.

NPV calculates the net present value of amounts to be received in the future, based on the interest rate you enter.

The number of cells in the range or list equals the number of periods (usually years), the value is computed over.

If you want to know the net present value of an investment which will return a single amount at some time in the future, you can use NPV, but you must be sure to enter zeros in the cells preceding the amount of the return, one cell for each period there will be no return. Then, include these cells in the range for NPV. For example, if the investment returns $5000 in four periods, the cells should contain 0, 0, 0, 5000.

---

## OR

`OR(`*`logical expression,logical expression`*`)`

The OR logical function lets you check to see if either or both of two logical expressions are true. If either or both expressions in its argument are true, the OR function is true. If both expressions are false, the OR function is false.

If you want to see if one and only one of two expressions is true, use the XOR (exclusive or) operator.

You can also use OR as a logical operator, as described in Part 1.

## PI

`PI`

PI has no argument. It lets you use the value of PI (the ratio of a circle's circumference to its diameter, approximately 3.14159) in a cell formula.

---

## PMT

`PMT(`*`p,i,n`*`)`

- *p* is the principal.
- *i* is the interest rate per period, in decimal form.
- *n* is the number of periods.

PMT calculates the payment due per period for a loan, based on the information you give it.

Be sure all the parts of the argument match. If you enter the length of the loan in months, be sure to enter the monthly (not annual) interest rate.

Each part of the argument can be any type of numeric expression, but all values must be positive. In addition, the interest rate must be between 0 and 1, and the number of periods must be an integer (no 3 1/2 month loans).

---

## PROPER$

`PROPER$(`*`string`*`)`

The PROPER$ string function converts the string in its argument to proper case (lower case with initial capitals). If `string` is a string constant, it must be in quotes.

You can change a string to all lower or upper case using LCS$ or UCS$.

## PV

`PV(`*`p,i,n`*`)`

- *p* is the payment per period.
- *i* is the interest rate per period, in decimal form.
- *n* is the number of periods.

PV is the present value function. It calculates the present value of an ordinary annuity based on the information you give it.

Be sure all the parts of the argument match. If the annuity has monthly payments, enter the monthly (not annual) interest rate, and the length of the annuity in months.

Each part of the argument can be any type of numeric expression, but all values must be positive. In addition, the interest rate must be between 0 and 1, and the number of periods must be an integer (no 3 1/2 month loans).

---

## RATE

`RATE(`*`endval,startval,periods`*`)`

RATE calculates the interest rate necessary for an investment of *startval* to increase in value to *endval* during the number of *periods* you specify.

The interest rate calculated will be for the length of one period—if you enter the number of years, the answer will be the yearly interest rate. RATE assumes that interest is compounded once per period, so *periods* should be the number of compounding periods for the investment.

RATE is similar to CTERM, which figures the number of periods necessary for an investment to increase at a specified interest rate.

## REG

`REG(`*`numeric expression,cell range`*`)`

- *`numeric expression`* is the value of X for which you want to find the Y value (see below).
- The *`cell range`* contains the pairs of XY values for which you want to find the regression equation. It must contain either two rows or two columns.

The regression function, REG, calculates the equation for the straight regression line describing the relationship between two sets of variables, X and Y. REG uses the method of least squares to find the equation, which has the form Y = aX + b. REG then uses this equation to find the value of Y for the value of X you specified.

*`Cell range`* must be in a single layer, and must be two cells wide in one direction (either two rows or two columns); it can be any length in the other. The first row or column contains all the values for the X variable; the second row or column contains the values for Y.

---

## RND

`RND(`*`number`*`)`

RND is AlphaCALC's random number generator. It generates a random number between 0 and 1, inclusive. You can then multiply the random number by whatever scale factor you need for the range of numbers you want.

The *`number`* in RND's argument can be any number. AlphaCALC uses it as a "seed" to generate the random number, but the number you use has no effect on the number generated.

---

## ROUND

`ROUND(`*`numeric expression,X`*`)`

ROUND rounds the value of *`numeric expression`* to *`X`* decimal places. This can help you avoid small discrepancies in totals caused by accumulated rounding errors.

ROUND differs from just setting the cell format to the desired number of decimal places—ROUND actually changes the value in the cell, not just the way it is displayed.

## ROWS

`ROWS(`*`cell range`*`)`

ROWS counts the number of rows in the cell range you enter.

---

## SGN

`SGN(`*`numeric expression`*`)`

The signum function, SGN, has a value of 1 if the value of its argument is positive, -1 is the argument is negative, or 0 if the argument equals 0. It can be useful when all you are concerned with is if a particular value is positive or negative.

---

## SIN and SIND

`SIN(`*`radians`*`)`
`SIND(`*`degrees`*`)`

SIN and SIND are AlphaCALC's two sine functions. If you have an angle expressed in radians, use SIN to find its sine; if the angle is in degrees, use SIND. Both *`radians`* and *`degrees`* can be any numeric expression.

You can use ASIN and ASIND to find the arcsine of a sine value.

---

## SLN

`SLN(`*`cost,salvage,life`*`)`

SLN is the simplest of the depreciation functions. It calculates the depreciation per period for an item that depreciates evenly over its useful life; this is called straight line depreciation.

Each part of the argument can be any type of numeric expression. Be sure to express *`life`* in the same units you want the depreciation value for (years for yearly depreciation, and so on).

For example, if an item cost $10,000 and depreciates evenly to a salvage of $1000 in ten years, its depreciation per year is: SLN(10000,1000,10) = 900.

DDB and SYD calculate depreciation values using other methods.

## SPACE$

`SPACE$(`*`numeric expression`*`)`

SPACE$ returns a string of blank spaces. The number of spaces depends on the value of the argument. You can use SPACE$ to help format report headings, or whenever you need a specific number of blank spaces.

The value of the argument for SPACE$ should be a positive integer.

---

## SQUARE ROOT

`SQRT(`*`numeric expression`*`)`
`SQR(`*`numeric expression`*`)`

The square root function calculates the positive square root of its argument. As shown, you can use either SQRT or SQR as the function name. If you use SQRT, AlphaCALC changes it to SQR.

Since negative numbers have no real square root, the value of the argument should not be negative.

---

## STDEV and STDEV1

`STDEV(`*`cell range or list`*`)`
`STDEV1(`*`cell range or list`*`)`

Both STDEV and STDEV1 compute the standard deviation (the square root of the variance) of the values of the cells in their arguments. The difference is the exact formula used.

STDEV uses the formula for an entire population, with N in the denominator.

STDEV1 uses the formula for data representing a sample of a population, with N-1 in the denominator.

You can calculate variances using VAR and VAR1.

## STR$

STR$(*expression*)

STR$ converts the expression in its argument, which may be a number, cell reference, or whatever, to a text string. You can then use the & operator or any function which takes a string argument with the result.

---

## SUM

SUM(*cell range or list*)

SUM adds the values of the cells in its argument and displays the total. It ignores text strings and logical values.

---

## SYD

SYD(*cost,salvage,life,period*)

The SYD function calculates an item's depreciation using the sum of years digits method. This is an accelerated depreciation formula—it depreciates an item faster during the early years of its life.

For SYD, you enter the item's original *cost*, its final *salvage* value, the length of its useful *life*, and the *period* for which you want its depreciation. Make sure the period you choose matches the units you use for the useful life; both are usually expressed in years.

For example, this function finds the depreciation during the third year for a $50,000 item which depreciates to $3000 in 20 years: SYD(50000,3000,20,3) = $4028.57

DDB and SLN are AlphaCALC's other depreciation functions.

---

## TAN and TAND

TAN(*radians*)
TAND(*degrees*)

TAN and TAND are AlphaCALC's tangent functions. If the angle is expressed in radians, use TAN to find the tangent; if the angle is in degrees, use TAND. Both *radians* and *degrees* can be any numeric expression.

The tangent of certain angles (in radians, PI/2, 3PI/2, etc.; in degrees, 90, 270, and so on) is undefined; if you try to find the tangent of one of these values, you receive an error message.

You can calculate arctangents using ATAN and ATAND.

## TIME

```
TIME
```

The TIME function, which has no argument, returns the current system time. The time is stored as the number of seconds since midnight; be sure to format the cell for the time display you want. The time formats are listed in Part 1.

## TODAY

```
TODAY
```

TODAY has no argument. It always returns the current serial date—the number of days between January 1, 1900 and today. You can use TODAY as part of the argument for other date functions or format the cell to print today's date as part of the worksheet. Available date formats are listed in Part 1.

When using TODAY, make sure your computer's system date is set correctly.

## TRUE

```
TRUE
```

The TRUE function does not take an argument—it always returns the value of TRUE.

TRUE can be very useful in checking the construction of formulas using other logical functions, especially complex IF functions. By substituting TRUE for the expression in an IF argument, you can make sure the result for the first (true) branch of the IF function is correct.

## UCS$

UCS$(*string*)

The UCS$ string function converts the string in its argument to all upper case. You then don't have to worry about typing the string in all capitals—AlphaCALC converts it for you.

If *string* is a string constant, it must be in quotes.

LCS$ and PROPER$ also change the case of strings.

---

## VAR and VAR1

VAR(*cell range or list*)
VAR1(*cell range or list*)

The two variance functions, VAR and VAR1, both calculate the variance of the values of the cells in their argument.

VAR calculates the variance using the formula for data describing an entire population, with N in the denominator.

VAR1 calculates the variance using the formula for data describing a sample of a population, with N-1 in the denominator.

You can find standard deviations using STDEV and STDEV1.

---

## VLOOKUP

VLOOKUP(*expression,cell range,offset*)

- *expression* is the string or numeric value you want to look for. If it is a string constant, it must be in quotation marks.
- *cell range* is the range containing the table you want to search.
- *offset* determines the column VLOOKUP returns the value from.

VLOOKUP performs a lookup function in a vertical table. It searches the left column of *cell range* for the value or string you're looking for. If there is no exact match for a numeric value, it finds the next lower value. Then, it moves to the right the number of columns specified in *offset*, and returns the value of the corresponding cell in that column.

For example, assume this simple table starts in cell A1:

```
2     WRENCH          $4.95
3     PLIERS          $1.99
5     SCREWS          $0.05
```

Here are some VLOOKUP functions and their results:

```
VLOOKUP(3,A1:C3,2)              $1.99
VLOOKUP("SCREWS",B1:C3,1)       $0.05
VLOOKUP(4,A1:C3,1)              PLIERS
```

Notice, in the last example, since there was no exact match for 4, VLOOKUP used the next lower value.

The values in the column you search must be in ascending order, and there should not be any duplicates. If there are duplicates, only the first occurrence is found.

HLOOKUP works on horizontal tables; you can use LOOKUP with any shape table.

## WEEKS

`WEEKS(`*`Date 1,Date 2`*`)`

WEEKS calculates the number of weeks between two dates. You can use this number to compute weekly average sales, weekly return on an investment, and so on.

Both dates are date strings. They must be in the correct format for your language (MM/DD/YY for American English). If they are literal strings, they must be enclosed in quotes.

You can use DAYS and MONTHS to find the number of days or months between two dates.

## YEAR

`YEAR(`*`serial date`*`)`

The YEAR function determines the calendar year for the serial date in its argument. Its value is a two digit number (the last two digits of the year).

You can extract just the day or month from a serial date using DAY or MONTH.

## YINT

`YINT(`*`P,I,N,MM/YY,YY`*`)`

- *P* is the principal.
- *I* is the monthly interest rate, expressed as a decimal.
- *N* is the length of the loan, in months.
- *MM/YY* is the starting month and year of the loan.
- *YY* is the year you want to know the interest for.

The YINT function calculates the interest due on a loan in a specific calendar year. This amount is often needed for tax purposes.

Each part of the argument can be any type of numeric expression, but all values must be positive. In addition, the interest rate must be between zero and one, and the number of months must be an integer (no 3 1/2 month loans).

When entering the argument, be sure to enter the monthly interest rate, not the yearly rate, and the loan length in months, not years.

# APPENDIX A
# COMMAND AND KEY LIST

| FEATURE | DESCRIPTION |
|---|---|
| AUTO ADVANCE | Advances cursor to next cell after entry. |
| AUTO INSERT | Changes formula entry to character insert mode. |
| BLOCK CLEAR | Clears block marks. |
| BLOCK COPY | Copies marked range to new location. |
| BLOCK ERASE | Erases unprotected cells in marked block. |
| BLOCK MARK | Marks range of cells—they can then be used in many other features. |
| BORDER COLUMN | Turns column border display on and off. |
| BORDER ROW | Turns row border display on and off. |
| COPY | Copies a worksheet. |
| CREATE | Creates a new worksheet. |
| CUT | Copies cell range into a new, empty worksheet. |
| DELETE COLUMN | Deletes current column, moves other columns to fill in. |
| DELETE ROW | Deletes current row, moves other rows to fill in. |
| DIRECTORY | Displays or prints list of worksheets in account. |
| DUPLICATE | Makes one or more copies of a range of cells. |
| EDIT | Calls up existing worksheet for viewing or changing. |
| EDIT CELL | Lets you change existing cell formula. |
| ERASE | Erases existing worksheet. |
| ERASE COLUMN | Erases current column, leaves blank. |
| ERASE ROW | Erases current row, leaves blank. |
| EXPORT | Creates other type of file from worksheet data. |
| FINISH | Writes worksheet to disk and exits. |
| FORMAT | Lets you change cell format. |
| GLOBAL | Searches for and replaces a group of characters. |
| GOTO CELL | Moves cursor to a specified cell. |
| GRAPH DISPLAY | Displays or prints graph. |
| GRAPH SETUP | Defines graph parameters. |
| HEADER | Lets you view and change descriptive and page format information for the worksheet. |
| HELP | Displays help information. |
| IMPORT | Takes specified format files and creates a worksheet or copies into existing worksheet. |
| INSERT COLUMN | Creates a blank column at cursor position. |
| INSERT ROW | Creates a blank row at cursor position. |

| FEATURE | DESCRIPTION |
|---|---|
| ITERATION | See "Recompute - iteration." |
| LOAN | Displays or prints loan amortization information. |
| LOCK COLUMN | Locks one or more columns on screen no matter where you move in worksheet. |
| LOCK ROW | Locks one or more rows on screen no matter where you move in worksheet. |
| MACROS | Lets you record and play back any series of keystrokes. |
| MODE | Makes temporary changes to operation settings. |
| MOVE | Lets you move one or more rows or columns to a new location in worksheet. |
| NEXT MATCH | During search, finds next occurrence of group of characters. |
| NEXT SCREEN | Goes to next screen of worksheet display. |
| NEXT WORD | Goes to next unprotected cell with data. |
| PAGE BREAKS | Inserts page break at cursor position. |
| PAGINATE | Shows where your worksheet will divide into pages when printed. |
| PASTE | Copies other worksheet into current worksheet. |
| PREV SCREEN | Goes to previous screen of worksheet display. |
| PREV WORD | Goes to previous unprotected cell with data. |
| PRINT | Prints all or part of worksheet. |
| PROTECT | Protects or removes protection from a range of cells. |
| PROTECT CELL | Protects or removes protection from a single cell. |
| QUEUE | Lets you view and modify the printer queue. |
| QUIT - Editing Menu | Leaves worksheet without writing to disk. |
| QUIT - Main Menu | Leaves AlphaCALC. |
| RECOMPUTE | Recalculates worksheet values. |
| RECOMPUTE - iteration | Recalculates circular references. |
| RENAME | Changes a worksheet's name. |
| REPLACE | Looks for and replaces a group of characters. |
| SAVE | Writes the worksheet to disk without leaving it. |
| SCREEN LEFT | Moves screen display left. |
| SCREEN RIGHT | Moves screen display right. |
| SEARCH | Looks for a group of characters in cell formula or value. |
| SORT | Sorts rows of worksheet based on values of specified column(s). |
| SPEED MODE | Makes [RETURN] and [ENTER] work like [EXECUTE] in cell formula field. |
| WIDTH | Changes worksheet display between 80 and 132 columns. |
| WRITE | Creates AlphaWRITE document from a worksheet. |

# APPENDIX B
# FUNCTION LIST

| FUNCTION | DESCRIPTION |
|---|---|
| ABS | Absolute value |
| ACOS and ACOSD | Arccosine (radians and degrees) |
| ACUM | Accumulator |
| AND | Are both expressions true? |
| ASIN and ASIND | Arcsine (radians and degrees) |
| ATAN and ATAND | Arctangent (radians and degrees) |
| AVERAGE | Compute average value |
| CMONTH | Convert month number to name |
| COLS | Count number of columns |
| COS and COSD | Cosine (radians and degrees) |
| COUNT | Count numeric values in a range |
| COUNTA | Count non-blank cells in a range |
| COUNTV | Count occurrences of a specific value |
| CTERM | Calculate time to maturity at a given interest rate |
| DATE | Return serial date for date |
| DATEVALUE | Return serial date for date |
| DAY | Calculate day of month for serial date |
| DAYS | Count days between two dates |
| DDB | Double declining balance depreciation |
| DING | Ring terminal bell |
| EXP | Calculate e to a given power |
| EXTCELL | Get value from another worksheet |
| FACT | Compute factorial |
| FALSE | Logical false |
| FISCALMONTH | Figure month of fiscal year |
| FIX | Truncate to an integer |
| FV | Calculate future value of an annuity |
| HLOOKUP | Horizontal table lookup |
| IF | Choose value based on truth/falsity of a logical expression |
| INDEX | Look up a cell within a table |
| INT | Round to largest integer less than or equal to value |
| LCS$ | Convert string to lower case |
| LN | Natural logarithm |
| LOG10 | Common logarithm |

| FUNCTION | DESCRIPTION |
|---|---|
| LOOKUP | Table lookup |
| MAX | Find maximum value |
| MIN | Find minimum value |
| MOD | Find remainder of a division |
| MONTH | Extract month number from a serial date |
| MONTHS | Count months between two dates |
| NOT | Reverse logical value |
| NPV | Net present value |
| OR | Is either expression true? |
| PI | The value of pi |
| PMT | Calculate payment per period for a loan |
| PROPER$ | Convert string into proper case |
| PV | Calculate present value of an annuity |
| RATE | Calculate interest rate for a given time to maturity |
| REG | Calculate regression value |
| RND | Generate random number from 0 to 1 |
| ROUND | Round to specified number of places |
| ROWS | Count number of rows in range |
| SGN | Is number positive, negative or zero? |
| SIN and SIND | Sine (radians and degrees) |
| SLN | Straight line depreciation |
| SPACE$ | Generate string of spaces |
| SQR | Square root |
| STDEV and STDEV1 | Standard deviation |
| STR$ | Convert value to text string |
| SUM | Add values in range |
| SYD | Sum of years digits depreciation |
| TAN and TAND | Tangent (radians and degrees) |
| TIME | Return current system time |
| TODAY | Return today’s date |
| TRUE | Logical true |
| UCS$ | Convert string to upper case |
| VAR and VAR1 | Variance |
| VLOOKUP | Vertical table lookup |
| WEEKS | Count weeks between two dates |
| YEAR | Calculate year for serial date |
| YINT | Calculate interest due in a specific year |

# APPENDIX C
# ERROR AND OTHER MESSAGES

This appendix lists the messages AlphaCALC may display to tell you the status of an operation.

The messages are in alphabetical order. Each message is followed by a short explanation, telling you what caused the message and what, if anything, you need to do about it.

These messages are AlphaCALC's way of communicating with you. Some of them tell you that AlphaCALC is performing or has completed an operation, so you know everything is working smoothly. Others inform you you've responded incorrectly to an AlphaCALC prompt, or that there's a potential problem with AlphaCALC or your computer.

Some messages you see during AlphaCALC are not generated by AlphaCALC. They are created by the operating system, AMOS, and merely displayed through AlphaCALC. These messages are not included here. If you see a message during AlphaCALC and cannot find it in this appendix, let your system operator know. It may indicate a problem with your operating system or computer which your system operator needs to correct.

### .GDV not found

This message may occur when you use the graph display feature. It indicates one of these two problems in the setup of the graphics feature:

- The graphics driver for the graphics device you chose is not loaded in either system or user memory.
- There are no graphics devices defined in the AlphaCALC initialization file for your account or the system GRFDEF initialization file.

The system operator can correct both of these problems. For more information on setting up the graphics feature, see the *AlphaCALC Installation and Administrator's Guide.*

### AlphaCALC file in old (unsupported) format

The AlphaCALC file you tried to use with **EDIT** is from a previous version of AlphaCALC. AlphaCALC automatically converts AlphaCALC 1.2 and 1.2A files, but worksheets from versions of AlphaCALC before 1.2 cannot be converted directly. You must convert these files to AlphaCALC 1.2 format, then convert them to 2.0.

You also receive this message if you try to use a pre-2.0 worksheet as a prototype worksheet in CREATE. You must convert a worksheet to 2.0 format before you can use it as a prototype.

### Amount difference entered is invalid

The loan amount difference you entered for the amortization analysis option of the LOAN command is illegal. The amount difference must be less than 10% of the base amount.

### AMSORT.SYS not found in memory

For the SORT command to work, the AMOS system sort program, AMSORT.SYS, must be loaded in system memory. If it isn't, you see this message when you use SORT.

The system operator can change your system initialization file to load AMSORT.SYS in system memory. See the *AlphaCALC Installation and Administrator's Guide* for details.

### Axis start value exceeds end value

When entering the Y axis values for your graph, the starting value you've entered is larger than the ending value. You need to correct these values for AlphaCALC to be able to create the graph.

### CALC.INI command obsolete - bypassing: *Command*

You may see this message when you call up AlphaCALC. The AlphaCALC initialization file for this account has a command in it which was valid for an earlier version of AlphaCALC, but no longer applies. *Command* is the obsolete command.

You'll receive this message for each obsolete command in the initialization file. These commands are ignored, and won't affect AlphaCALC's operation in any way. You can remove them from the initialization file if you want to stop the message from appearing.

### Can't shift non-blank cell off sheet

You're trying to insert a row or column in your worksheet, and doing so would move a cell with a cell formula in it off the edge of the current layer—past row 8191 or column IU. This is not allowed.

You can avoid this message by rearranging the current layer to fit all the information you want, or by moving some cells to another layer of your worksheet.

### Cannot directly EXPORT 1.x format worksheets

In the **EXPORT** command, you've entered the name of an AlphaCALC 1.2A or earlier worksheet. Before you can **EXPORT** this worksheet from AlphaCALC 2.0, you must convert it to 2.0 format by editing and saving it.

### Cannot edit/modify - cell is protected

You tried to change the cell formula in a protected cell. You cannot edit or erase a protected cell.

You can tell if a cell is protected by looking at the cell ID at the bottom left of your screen. If the current cell is protected, the cell ID is in brackets.

If you need to change a protected cell, use the [PROTECT CELL] key or **PROTECT** command to remove protection from the cell.

### Cell *Cell ID*

This message at the lower left of your worksheet display shows the cell ID of the current cell.

### Cells on other layers may be affected - [EXEC] to proceed; [MENU] to cancel

You are using a feature, such as **DUPLICATE** or [BLOCK ERASE], which will affect cells on more than one layer. It is perfectly allowable to do this, but since you may erase or copy over cells you can't see on your screen, AlphaCALC displays this warning message and asks you to confirm the operation.

To perform the operation, press [EXECUTE]; to cancel the request, press [MENU].

If you don't want to see this warning message when you perform multiple layer operations, you can turn it off using an AlphaCALC initialization file command. See the *AlphaCALC Installation and Administrator's Guide* for more information.

### Characters per line is too long

The characters per line field of the Worksheet Header only accepts values up to 255. If you want to print more than 255 characters per line, you must print by cell range, not by page.

### Circular reference detected

This message may appear any time AlphaCALC is calculating a cell formula—after you enter a cell formula, during [RECOMPUTE], and so on. One of your cell formulas directly or indirectly refers to itself. For example, cell A4 may refer to F3, which has a formula including a reference back to A4.

If you intentionally use a circular reference, you can use the [ITERATION] key to compute its value. See the section on "Recompute with Iteration" in Part 2.

### Data area is full

The graphics feature must be able to store all the data for the graph in your memory partition. The area of memory set aside for this is not large enough for this graph.

The system administrator can change the amount of memory reserved for graphics by using the **MODE** command or the MAX-GRAPH-POINTS command in the AlphaCALC initialization file. See the *AlphaCALC Installation and Administrator's Guide.*

### Delta cell has an invalid value type

The cell you've chosen as the delta cell during the recompute iteration option does not have a numeric value. Only numeric values can be checked for the delta change value.

### Delta cell has no value

The cell you've chosen as the delta cell during the recompute iteration option is blank. The delta cell must have a numeric value.

### Delta value must be more than zero

You cannot use zero as the delta amount to check for in the recompute with iteration feature. The delta amount must be positive.

### Destination range overlaps source range

The destination range for **DUPLICATE** cannot overlap any part of the source range. If you need to copy a range to a location which partially overlaps itself, first copy the cells to another area of your worksheet, then copy that range to the desired final destination.

### DIF format error in line: *XXX*

The DIF file you're trying to IMPORT has a format error in line *XXX*. The IMPORT stops with the line before the error. If you're familiar with DIF format, you can use AlphaVUE to look at the file and correct the line.

For a description of DIF format, see the *AlphaCALC Programmer's Information Manual*.

### Division by zero

One of your cell formulas divides by zero at some point. Since this is an impossible operation, the value for the cell, and any cells that refer to it, are not correct.

Because of the way math errors are handled internally, a cell with this error may display the largest or smallest value the computer is capable of handling. This value may vary depending on your hardware and your version of the operating system.

### Error unpacking pre-2.0 expression in cell XXX

While converting a worksheet to 2.0 format, AlphaCALC has found a cell formula it cannot convert properly. The formula is changed to a text entry. Note the cell ID shown, then after the worksheet is displayed, edit the cell to correct the formula.

### Expression is too long

You're replacing a string with a longer string in REPLACE or GLOBAL, and the replacement would cause a cell formula to be longer than 254 characters. The cell formula is not changed. The string is still replaced in other cells.

### Expression parentheses are unmatched

In entering your formula, you have more left parentheses than right parentheses. For AlphaCALC to know how to evaluate your formula, the parentheses must match.

### Expression syntax is invalid

AlphaCALC does not understand your cell formula. Check for typing errors, and make sure all function arguments are in the proper format and enclosed in parentheses. Also, make sure any string constants in the formula are in quotes.

After you press a key to clear the error message, the cursor goes to the part of the formula AlphaCALC doesn't understand.

### Fatal error - CALC memory corrupted

An internal error has corrupted the information AlphaCALC has stored in user memory. You are returned to AMOS command level, and your worksheet returns to the state it was in when you last used SAVE or FINISH.

Report this error to your system operator.

### Field overflow in line: *XXX*

The file you're importing contains too much information on line *XXX*. IMPORT skips this line and continues importing the rest of the data. If you are familiar with the import file format, you can use AlphaVUE to edit the file and try to correct the line, then use IMPORT again.

See the *AlphaCALC Programmer's Information Manual* for descriptions of import file format.

### File format error. Some data may be lost.

The file you're importing contains data in a format AlphaCALC doesn't understand. IMPORT skips the offending line, then continues importing data. If you are familiar with the import file format, you can use AlphaVUE to edit the file and try to correct the problem, then use IMPORT again.

See the *AlphaCALC Programmer's Information Manual* for descriptions of import file format.

### File is not AlphaCALC compatible

You tried to call up a worksheet, but the file, even though it has an .ALC extension, is not an AlphaCALC worksheet. Only AlphaCALC worksheet files should have the .ALC extension.

You can also get this message if you try to PASTE an AlphaCALC worksheet that hasn't been converted to 2.0 format. You must convert worksheets using EDIT before you can use them with other commands.

It is also possible to get this error if system problems have corrupted one of your worksheet files. If you think this is the case, contact your system operator.

### Floating point overflow

A calculation's result is larger than the largest value AlphaCALC can handle—this is about 1.7E38. AlphaCALC assigns this largest value to the calculation.

### Floating point underflow

A calculation's result is smaller than the smallest value AlphaCALC can handle—this is about 1.7E-38. AlphaCALC assigns this smallest value to the calculation.

### Format cannot support multiple layers.

You are trying to **EXPORT** a multiple layer worksheet to a file type which doesn't support multiple layers. Only the Cell ID, Value format supports multiple layers. Either use this format or **EXPORT** the worksheet one layer at a time.

### GRAPH.SYS not found.

The GRAPH.SYS file must be loaded into system memory for the graphics feature to work. It is possible the AMIGOS package, which is necessary for graphics, has not been installed on your system. Contact your system operator.

See the *AlphaCALC Installation and Administrator's Guide* for information on installing graphics.

### I'm sorry, there is no further help available

You've pressed the HELP key at a time when there is no more help available.

### Illegal cell contents encountered in line: *XXX*

The DIF or SYLK file you're importing contains illegal information on line *XXX*. **IMPORT** skips the cell with the error and continues importing data. If you are familiar with the import file format, you can use AlphaVUE to edit the file and try to correct the line, then use **IMPORT** again.

See the *AlphaCALC Programmer's Information Manual* for a description of import file formats.

### Illegal command in CALC.INI file: *Command*

There is a command in your AlphaCALC initialization file which AlphaCALC does not understand. You may have typed a command incorrectly, or forgotten to put a semicolon (;) before a line which is supposed to be a comment.

This is an information only message; the unrecognized line is ignored and AlphaCALC will function correctly.

### Illegal field type descriptor data in line: *XXX*

The SYLK file you're importing contains illegal information on line *XXX*. **IMPORT** skips the cell with the error and continues importing data. If you are familiar with the import file format, you can use AlphaVUE to edit the file and try to correct the line, then use **IMPORT** again.

See the *AlphaCALC Programmer's Information Manual* for a description of import file formats.

### Incompatible SSD overlay

The different software modules making up AlphaCALC are not compatible. Contact your system operator.

System Operator: Run VERIFY to check the versions of the AlphaCALC software. If necessary, re-install AlphaCALC or restore it from a backup tape. See the *AlphaCALC Installation and Administrator's Guide* for information on using VERIFY.

### Incompatible Value Types

Your cell formula is trying to combine or compare two values of different types, which AlphaCALC can't compare. For example, you may be trying to compare a logical value and a string using AND or OR.

### Incompatible version of AMSORT.SYS

The **SORT** command uses the AMOS sorting routine, AMSORT.SYS. The version of AMSORT.SYS on your computer is not compatible with AlphaCALC. Contact your system operator.

System Operator: Versions of AMSORT.SYS for both AMOS 1.X and AMOS 2.X systems are included with the AlphaCALC software. Run VERIFY to make sure these files were copied to ALC: correctly, and make sure you copied the correct version of AMSORT.SYS to the SYS: account. See the *AlphaCALC Installation and Administrator's Guide*.

### Insufficient free memory available

There is not enough room in your memory partition to perform the operation you requested. Contact your system operator.

System Operator: if possible, increase the memory partition for this user. Refer to the chapter on AlphaCALC memory usage in the *AlphaCALC Installation and Administrator's Guide.*

**Insufficient memory available to SORT**
**Insufficient Tool Box memory pool resources - Contact System Operator**

The SORT command requires memory in excess of that needed to load your worksheet in order to store data during the sort. There is not enough room in your memory partition for SORT to do this. Contact your system operator.

System Operator: If possible, increase the memory partition size or the toolbox memory pool size. See the *AlphaCALC Installation and Administrator's Guide* chapter on memory usage. If this isn't possible, this user will have to use smaller worksheets or use another job, with more memory, for sorting.

**Interest rate difference entered is invalid**

The interest rate difference you entered for an amortization analysis is not in the acceptable range. The difference must be no more than 1% of the base interest rate.

**Interest rate entered is invalid**

The interest rate you entered for an amortization analysis is not in the acceptable range. The interest rate must be less than 99.99%, and entered as a percentage, not a decimal (enter 12 for 12%, not .12).

**Invalid Auto-Advance selection**

The auto advance direction entry must be a three letter code, consisting of one letter from each of these pairs: R or L, U or D, and I or O. Any other entry is invalid. See "Auto Advance" in Part 2.

**Invalid cell expression in cell XXX**

As AlphaCALC is loading a worksheet, it detected an illegal cell entry. This may mean your file has been corrupted on the disk. Contact your system operator.

**Invalid cell range specification.**

Your entry isn't in the correct format for a cell range. Type your entry again, with the correct format. Cell ID and cell range format are discussed in Part 1.

You can also receive this message is you press [BLOCK COPY] or [BLOCK ERASE] with no block marked.

**Invalid cell-ID in line: *XXX***

The file you're importing contains an invalid cell ID on line *XXX*. If you are familiar with the import file format, you can use AlphaVUE to edit the file and try to correct the line, then use **IMPORT** again.

Import file formats are described in the *AlphaCALC Programmer's Information Manual*.

**Invalid cell-ID specification**

The cell ID you entered is not in the correct format, or is greater than IU|IU8191. Check for typographical errors and try again. Cell ID format is described in Part 1.

**Invalid date entered**

The date you've entered is illegal. For example, you may have entered 12/41/91 instead of 12/31/91.

**Invalid date format**

You have entered a date in the incorrect format. Dates must be entered in the correct format for your currently selected language—for American English, this is MM/DD/YY format. Check for typos and try entering the date again.

**Invalid expression argument**

The argument you've entered for an AlphaCALC function is not in the right format. If you're not sure what is wrong with the argument you've entered, check the function's description in Part 3 for the correct argument format.

**Invalid iteration count**

The iteration count you entered is not in the acceptable range, 1 to 255.

### Invalid MOVE command

To use the MOVE key, these conditions must be met:

- There must be a marked block consisting only of cells in row 1 (to move columns) or cells in column A (to move rows).
- If you're moving rows, the cursor must be in column A. If you're moving columns, the cursor must be in row 1.
- You can't move a row or column to another layer.

If you press MOVE without a correctly marked block, or with the cursor in the wrong position, you receive this message.

### Invalid sort key

Your entry in one of the sort key fields is illegal. Either you didn't enter a column letter, or the column you entered is not in the sort range.

### Invalid Value Type

In a cell formula, you're trying to perform an operation which isn't allowed for the type of value you're using. For example, you may be trying to add a logical value and a number.

### Maximum Number of Simultaneous Users = XX

On each computer where it's installed, AlphaCALC is licensed for a particular maximum number of users. Anyone who tries to access AlphaCALC while the maximum number of people are already using it will receive this message. You'll have to wait until someone leaves AlphaCALC before you can use it.

### Negative values not allowed in specified graph

Negative values cannot be graphed in a pie chart. To graph negative values, you'll have to choose a different type of graph.

### No GRAPH memory available

There is not enough room in your memory partition for you to use the graphics feature. Decreasing the MAX-GRAPH-POINTS value using the **MODE** command may fix this problem. If it doesn't, contact your system operator.

System Operator: For a discussion of the graphics feature's memory requirements, and graph setup in general, see the *AlphaCALC Installation and Administrator's Guide.*

### No block marked

You've tried to copy or erase a block of cells, or clear block marks, and there is no block of cells marked. You muse use [BLOCK MARK] to mark the cells you want before using any of these features.

### No more matches found

In either the **SEARCH** or **REPLACE** command, you've passed the last occurrence of the search string in your worksheet.

### No replacement string specified

You must enter a replacement string in **REPLACE** or **GLOBAL** before you press [EXECUTE]. Type the string you want to replace the search string with, then press [EXECUTE] again.

### No search string specified

You must enter a string to search for in **SEARCH**, **REPLACE** or **GLOBAL** before you press [EXECUTE]. Type the string you want to look for, then press [EXECUTE] again.

You will also see this message if you press [NEXT MATCH] without having used the **SEARCH** command.

### Number of months entered is invalid

The loan length you entered is not in the legal range of 1 to 998 months. Also, it must be an integer; you cannot use partial months in the **LOAN** command.

### Parameter area is not large enough.

There is not enough room in your memory partition for you to use the graphics feature. Decreasing the MAX-GRAPH-POINTS value using the **MODE** command may fix this problem. If it doesn't, contact your system operator.

System Operator: For a discussion of the graphics feature's memory requirements, and graph setup in general, see the *AlphaCALC Installation and Administrator's Guide.*

### * * Please Contact Your System Operator * *

When you see any message that includes these words, there could be a problem with AlphaCALC's setup, or a potentially more serious problem with your computer. Let your system operator know what happened immediately.

**Printer driver not available**

The printer driver for the printer you've requested is not in system or user memory. This usually indicates a problem with the printer setup during AlphaCALC installation. Contact your system operator.

**Product Installation Code = *XXX***

You will see this message, where *XXX* is the PIC for AlphaCALC on your computer, when you use the CALC/P command to display the PIC. It is not an error message.

**Queue request denied - file not from your system**

In the QUEUE command, you've tried to delete or change the order of a file that's printing from another computer connected to yours via a network. You can affect only files printing from your own computer system with QUEUE.

**Requested data range label has no data point stored.**

The cells you've defined as a data range for the graph do not have any data in them, so there are no values for the graph to use. Check to make sure you entered the correct cell range.

**Requested file is no longer found in printer queue**

In the QUEUE command, you've tried to delete or change the order of a file which is no longer in the queue. The file may have finished printing, or it may have been deleted by someone else after you invoked the QUEUE command.

**Requested printer not found**

AlphaCALC cannot communicate with the printer you've selected. This could indicate a problem with the software controlling the printers on your computer. Contact your system operator.

**Search concluded**

In either the SEARCH or REPLACE command, you've passed the last occurrence of the search string in your worksheet.

**SORT error**

An internal error has occurred during the SORT command. If this error occurs repeatedly, it could indicate a serious problem with your computer. Contact your system operator.

### Specified string was not found

The string you entered in SEARCH, REPLACE or GLOBAL cannot be found anywhere in the cell range you're searching. Check to make sure:

- You entered the string to look for correctly.
- You're searching the cell range you want.
- You selected the proper type of cell formulas or values.
- In SEARCH, did you specify a complete match only?

### SYLK format error in line: *XXX*

The SYLK file you're importing contains a format error on line *XXX*. If you are familiar with SYLK file format, you can use AlphaVUE to edit the file and try to correct the line, then use IMPORT again.

### The item you selected is not recognized - please try again

AlphaCALC doesn't recognize your entry on a menu or command screen. Check for typing errors and try again.

### The password you entered does not match

You're trying to access a password protected worksheet, and you've entered the password incorrectly. If you think the password is correct, try again, being sure to avoid typing errors.

### Unrecognized .INI file command

There is a problem with AlphaCALC's setup. Contact your system operator.

System Operator: This message indicates a problem in the ESP toolbox initialization file, DMSESP: TOOLBX.INI, which is used by AlphaCALC. Check this file for errors.

### Unrecognized keyword in .GDF file

The parameters for an AlphaCALC graph are stored in a a graph definition file with the extension .GDF. The .GDF file for your graph has data in it which AlphaCALC does not understand. It's possible someone has manually edited the .GDF file.

If you are familiar with .GDF file format, as described in the AMIGOS/GRAPH documentation available from Alpha Micro, you can edit the file using AlphaVUE and fix the offending keyword. If you don't know how to do this, you may need to delete the .GDF file and redefine your graph.

**User impure area unavailable**

This message may be caused by an error in the DMSESP:TOOLBX.INI file. If you've set the USERS statement in this file to too small a number, the toolbox does not reserve enough memory for all AlphaCALC users. See the *AlphaCALC Installation and Administrator's Guide* chapter on memory usage.

**Warning: Data after END statement is ignored.**

The SYLK file you're importing contains information after its END statement. Any information after END is not imported. If you're familiar with SYLK format, you can use AlphaVUE to edit the file and IMPORT it again.

**Worksheet file NAME has been copied**

The COPY command displays this message to let you know the worksheet you requested has been successfully copied.

**Worksheet file NAME has been erased**

The ERASE command displays this message to let you know the worksheet you requested has been successfully deleted.

**Worksheet file NAME has been renamed**

The RENAME command displays this message to let you know the worksheet you requested has been successfully renamed.

**Your terminal does not support changing of display width**

You can use the WIDTH command only if your terminal supports both 80 and 132 column display widths. If you use WIDTH on a terminal which doesn't support both widths, you see this message.

TOC// APPD.WRT
COMTOC// AlphaCALC Reference Book

# APPENDIX D
# UNIVERSAL KEY SEQUENCES

The following table shows the universal key sequence to use for any AlphaCALC key feature. Features are listed alphabetically by AlphaCALC feature name.

These sequences will work on any terminal type Alpha Micro supports. For most terminals, you can invoke these features more easily using the function keys or special keys. Keyboard overlays showing key assignments for the various terminals are available from Alpha Micro. Also, see the *AlphaCALC Keyboard Command Quick Reference Card* for a complete list of key assignments for Alpha Micro supplied terminals.

## TO ENTER A KEYBOARD COMMAND

If two or more keys are shown, separated by spaces, press the first key, then press the second. These sequences often include the [ESCAPE] (or [ESC]) key. For example, if a key sequence is ESCAPE Q, press [ESCAPE] once, then type Q.

If two or more keys are shown, separated by hyphens, press both keys simultaneously. This is easiest to do by holding down the first key, pressing the second, then releasing both. For example, if the sequence is CTRL-P, hold down [CTRL] and type P once.

### A NOTE ON CASE

For all sequences including letter keys:

- If the letter comes immediately after the [ESC] key, you must type the letter in the case shown. Typing a lower case letter instead of a capital, or vice-versa, will not perform the feature you want.
- For all other sequences, such as those using the [CTRL] key, you can use either upper or lower case letters.

### IF YOU DON’T HAVE AN ESCAPE KEY

If your terminal does not have an [ESCAPE] key, use the key sequence [CTRL] / [[] (left square bracket) whenever the [ESCAPE] key is called for in the table.

| FEATURE | UNIVERSAL KEY SEQUENCE |
|---|---|
| Auto Advance | CTRL-\ |
| Auto Character Insert | CTRL-Q |
| Beginning of Layer | CTRL-^ |
| Beginning of Row | CTRL-U |
| Block Clear | ESCAPE Q |
| Block Copy | ESCAPE C |
| Block Erase | ESCAPE D |
| Block Mark | CTRL-P |
| Bottom of Screen | ESCAPE CTRL-J |
| Bottom of Layer | CTRL-E |
| Cancel | ESCAPE ESCAPE |
| Center Screen | CTRL-S |
| Character Delete | CTRL-D |
| Character Insert | CTRL-F |
| Column Delete | ESCAPE CTRL-D |
| Column Insert | ESCAPE CTRL-F |
| Cursor Down | CTRL-J |
| Cursor Left | CTRL-H |
| Cursor Right | CTRL-L |
| Cursor Up | CTRL-K |
| Display Column Border | ESCAPE B |
| Display Row Border | ESCAPE b |
| Edit Cell | ESCAPE E |
| End of Row | CTRL-N |
| Erase Column | ESCAPE CTRL-V |
| Erase Row | ESCAPE CTRL-Y |
| Execute | ESCAPE X |
| Goto Cell | CTRL-G |
| Graph Display | ESCAPE G |
| Graph Setup | ESCAPE g |
| Help | ESCAPE ? |
| Lock Column | ESCAPE L |
| Lock Row | ESCAPE l (lower case L) |
| Macros | ESCAPE @ |
| Menu | ESCAPE ESCAPE |
| Move Row or Column | ESCAPE M |
| Next Cell | CTRL-W |
| Next Field (while editing cell) | TAB |
| Next Layer | ESCAPE I |
| Next Match | CTRL-X |
| Next Screen | CTRL-T |
| Previous Cell | CTRL-A |
| Previous Field (while editing cell) | CTRL-] |
| Previous Layer | ESCAPE O |
| Previous Screen | CTRL-R |
| Protect Cell | ESCAPE p |
| Recompute | ESCAPE R |
| Recompute Iteration Option | ESCAPE r |
| Row Delete | CTRL-Z |
| Row Insert | CTRL-B |
| Screen Left | ESCAPE [ |
| Screen Right | ESCAPE ] |
| Speed Mode | ESCAPE CTRL-S |
| Top of Screen | ESCAPE CTRL-K |

# DOCUMENT HISTORY

| REVISION | DATE | DESCRIPTION |
|---|---|---|
| 00 | 10/3/91 | New document for AlphaCALC 2.0 release. Replaces old *AlphaCALC Reference Guide* and *AlphaCALC Reference Booklet*. |

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